NBCUniversal, Skypath and Amazon Web Services

L. MacLearn v0.7, 1/25/23

Contents

Table of Figures

Overview

Currently, NBCUniversal uses Amazon Web Service (AWS) to deliver IT innovation, reduce costs, and expand the reach of its local news stations. AWS provides NBCUniversal with the infrastructure and services needed to support its advanced targeting solutions, local news streaming channels, and SAP Journey to the Cloud, while defining and maintaining secure affiliate access with AWS Identity Access Management.

In 2023, NBCUniversal is increasing its commitment to further integrating Amazon Web Services (AWS) in support of and advancing NBCU's technology initiatives in several ways:

- **Advanced targeting solutions**: In February 2023, NBCUniversal announced new advanced targeting solutions and self-service technology, with key partners including AWS1.
- **Local news streaming channels**: In May 2023, NBCUniversal and Amazon will announce a partnership to expand the reach of NBC and Telemundo local news stations in streaming, with new FAST (free ad-supported streaming TV) channels presenting all-day coverage from NBC and Telemundo stations from the top TV regions. NBCUniversal used AWS services to facilitate the most live streamed Super Bowl and Olympics in history2.
- **SAP Journey to the Cloud**: NBCUniversal's SAP Journey to the Cloud was documented in a video by AWS, which explains how NBCUniversal used AWS to deliver IT innovation globally while reducing costs.
- **Increasing IT Security with implementation of OAuth 2.0**: NBCUniversal has introduced OAuth 2.0 in order to enable users to authorize third-party access to their servers without handing out their username and password. This capability provides Distribution Engineering the ability to remotely secure designated access, employing tokens to decouple unnecessary authentication data from authorization for program delivery and communication between NBCU and its affiliates, while supporting varied use cases according to the varying needs and device capabilities of those affiliates.
- **Providing secure access to NBCUniversal's affiliates**: NBCUniversal is leveraging AWS IAM to create a secure and scalable system for managing affiliate station access to their program distribution platform. Here's how it works:
  - User Directory: NBCUniversal uses Amazon Cognito to create a secure user directory for all affiliate stations. Each station would have its own unique user account.
  - IAM Roles: Different IAM roles could be created based on the type of affiliate (e.g., local NBC station, Telemundo affiliate) and their specific access needs.
  - Access Policies: NBCUniversal defines granular access policies using IAM. These policies specify which programs and resources each affiliate role can access, and what actions they can perform (e.g., view, download, stream).

- Multi-Factor Authentication: For enhanced security, NBCUniversal implements multi-factor authentication using Amazon Cognito for all affiliate users accessing sensitive content.
- Federated Access: For larger affiliate groups with their own identity systems, NBCUniversal uses *IAM Identity Center* to enable federated access, allowing affiliates to use their existing credentials.
- Temporary Credentials: When affiliates need to access specific AWS resources (like S3 buckets containing program files), NBCUniversal uses IAM to issue temporary, limited-privilege security credentials.
- Access Monitoring: NBCUniversal uses *IAM Access Analyzer* to continuously monitor and adjust access permissions, ensuring affiliates only have the necessary access.
- Integration with Content Delivery: The IAM system is integrated with NBCUniversal's content delivery network, ensuring that only authenticated and authorized affiliates can access and distribute specific programs.
- Automated Provisioning: When new affiliate stations are onboarded, NBCUniversal uses AWS IAM APIs to automatically create the necessary user accounts and assign appropriate roles and permissions.
- Compliance and Auditing: IAM provides detailed logs of all access attempts and changes to permissions, helping NBCUniversal meet compliance requirements and conduct regular audits of affiliate access.
- This system allows NBCUniversal to securely manage access for hundreds or thousands of affiliate stations, ensuring that each station has the right level of access to the programs they're authorized to distribute, while maintaining tight control over their valuable content.

This system would allow NBCUniversal to securely manage access for hundreds or thousands of affiliate stations, ensuring that each station has the right level of access to the programs they're authorized to distribute, while maintaining tight control over their valuable content.

AWS Data Lake for Programming Storage

NBCUniversal employs AWS Data Lake to centralize and secure its vast library of program streams and scheduling information for affiliate distribution. This system, referred to internally as NBCUniversal Content Hub (NCH), leverages various AWS services to create a scalable, secure, and efficient content management platform.

Key components of the NCH include:

- Data Storage: *Amazon S3* serves as the core storage layer, housing petabytes of video content, metadata, and scheduling information. Different S3 buckets are used to separate content based on type, sensitivity, and access requirements.

- Data Catalog: *AWS Glue* is used to create and maintain a comprehensive catalog of all content and metadata, making it easy to search and retrieve specific programs or scheduling information.
- Access Control: AWS Lake Formation is implemented to manage fine-grained access controls, ensuring that affiliates can only access the content and scheduling information they're authorized to distribute.
- Data Processing: Amazon EMR is used for large-scale data processing tasks, such as transcoding video content for different delivery formats or analyzing viewership data.
- Content Delivery: Amazon CloudFront is integrated to efficiently deliver content to affiliates worldwide, ensuring low-latency access to program streams.
- Security: *AWS Key Management Service* (KMS) is used for encryption of sensitive data, while AWS Identity and Access Management (IAM) manages user authentication and authorization.
- Analytics: Amazon Athena is used to run ad-hoc queries on the data lake, allowing NBCUniversal to gain insights into content performance and affiliate usage patterns.

NCH Workflow:

An NBCUniversal content team uploads a new program to the Content Hub via a secure web interface. The video is automatically processed by AWS Elemental *MediaConvert* to create multiple formats for different delivery channels. *AWS Glue* catalogs the new content, extracting metadata and linking it to relevant scheduling information. Affiliate stations access the Content Hub through a secure portal, authenticated via *AWS Cognito*.

Based on their access permissions (managed by Lake Formation), affiliates can view available content, download scheduling information, and stream programs directly from the platform. All access and usage is logged for auditing and analytics purposes.

Finally, NBCUniversal uses Amazon *QuickSight* to create dashboards showing content popularity, current ratings data, affiliate engagement, and other key metrics.

Engineering Overview

This section provides several engineering-specific topological views and descriptions of the system, including:

- A High-Level Conceptual view of the end-to-end workflow
- Channel origination (Zone 1)
- The Compression System (Zone 2 & 3)
- Satellite Uplink (Zone 4)
- IRD Downlink Behavior (Zone 5)

High-Level End-to-End Conceptual View

This conceptual view shows the components as they fit into the overall distribution ecosystem. Program outputs from our release chains will feed into the Electra XOS encoders, which are muxed together by the Prostream (PSXVM) units before sent to our existing uplink sites. The Decoder Management System, (DMS) will interface with Prostream multiplexers to add encryption and IRD management, and NMX will talk to all components for management and control. Out of scope for this Compression project is the deployment of the Harmonic 7721 IRDs at each affiliate, but this conceptual shows those components to illustrate the end-to-end workflow.

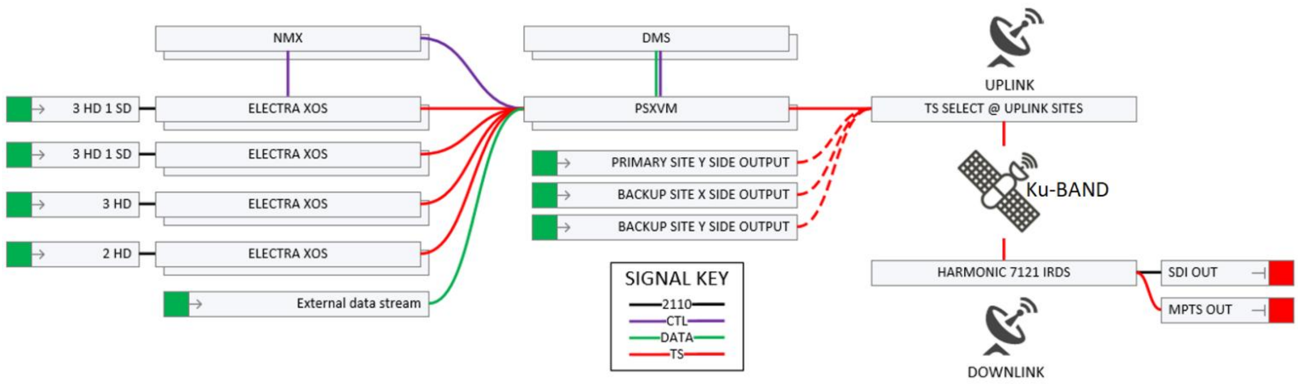


Figure 1 - Ku Band Distribution Ecosystem

The high-level view comprises five functional zones (numbered 1-5), shown in the next figure. These include:

- Zone #1 - Release Chain and 2110 Gateways
- Zone #2 – Electra XOS Encoders
- Zone #3 – Prostream X Multiplexers
- Zone #4 - DMS & IRDs
- Zone #5 – Uplink Sites

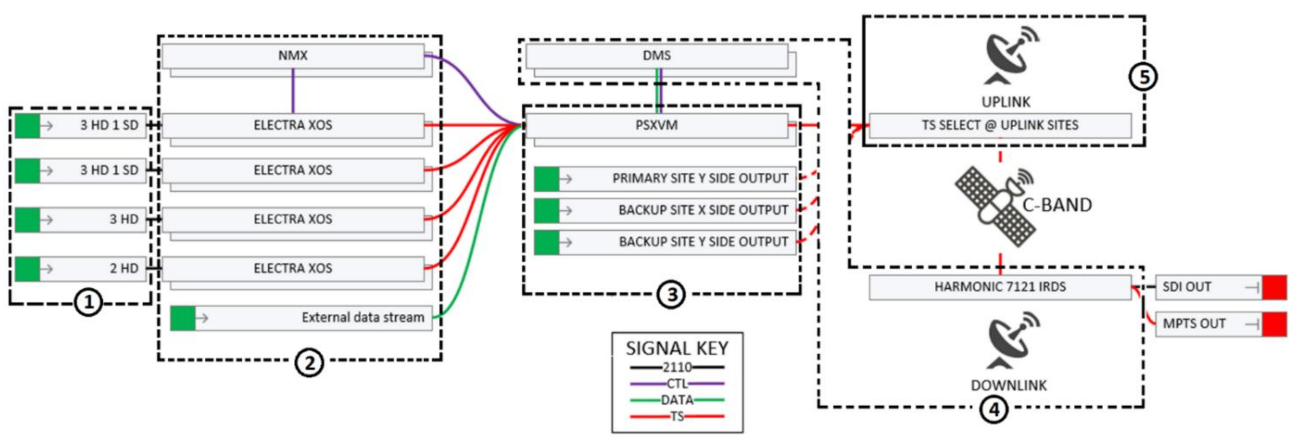


Figure 2 - Ku Band Distribution Ecosystem Zones

Compression Overview

NBC channel chains feeds into IP Gateways (IPG's) which presents those ready for air source streams on the Spine Leaf network as SMPTE 2110 IP video. Those video streams flow through an A and B leaf to our encoders which perform a 2022-7 *hitless merge* process that protects against network interruptions.

The encoders process those inputs and create the component streams that will be combined into our final statmux by the Prostream multiplexer servers. That Prostream-to-Encoder communication will be handled on a dedicated 10G data switch, which isolates that 'noisy' traffic on a dedicated 'island' switch. An external data stream will be generated by an LRM service and will be integrated into the mux on that same island switch. The DMS system will interface with the Prostream to encrypt and insert IRD data. The Prostream will create the MPTS statmux and present it back out to the Spine Leaf network for distribution as a 2022-7 compliant MPEG2 Transport stream with RTP headers.

Management of the ecosystem is handled by the NMX servers. In this ecosystem, each critical component has a backup host, and the NMX system is the interface to that orchestrates that failover process. NMX is also the operator interface for the commissioning team, as well as the API interface for our management and control system (DataMiner).

The ready for air mux outputs will flow through our existing WAN distribution paths to uplink sites with minimal modification. Additional X/Y and Primary/DR switching happens at the uplink site to further protect the on-air product.

Physical/Virtual Architecture

All Skypath Harmonic systems are virtualized on Cisco UCS C220 servers. The encoders are 'bare metal', one encoder per C220. The Prostream X units are installed on a Linux instance along with NMX (Windows 2019 server) and DMS (Windows 2019 Server). In this case several components, installed on a C220, are running VMware ESXI virtual machines.

Compression Conceptual View (Zones 1-3)

This conceptual view shows how the components of the Skypath compression system are connected for a single (X or Y) implementation at a single site. All switch and server hardware is duplicated for a true X/Y environment at each site. Management/Control connections are shown in **GREEN**, 2110 source feeds and Ready For Air (RFA) mux outputs are shown in **RED** and **BLUE** (A and B 2022-7). Mux elementary streams, Data PID stream, and statmux reflexing shown in **Orange**. The Distribution Muxes, show in **BLACK**, are passed through WAN gateways to our uplink sites.

Figure 3 (below) provides the physical connectivity between components but does not clearly show the signal flow through the ecosystem. Figure 4 (following page) illustrates the flow of services through the ecosystem.

2110 inputs (BLUE) feed Spine Leaf and flow to encoders, which encode Mux Component streams to the data switch. The statmux servers pull Mux Streams (ORANGE) and the Data Stream (ORANGE) in and apply DMS encryption before sending the Mux Outputs (RED) back to the Spine Leaf. A Nevion Virtuoso unit is used to locally switch between redundant Mux Outputs and provide a single Ready for Air Mux Output (GREEN). A WAN Gateway device is then responsible for sending that Mux Output across the WAN to our uplink sites.

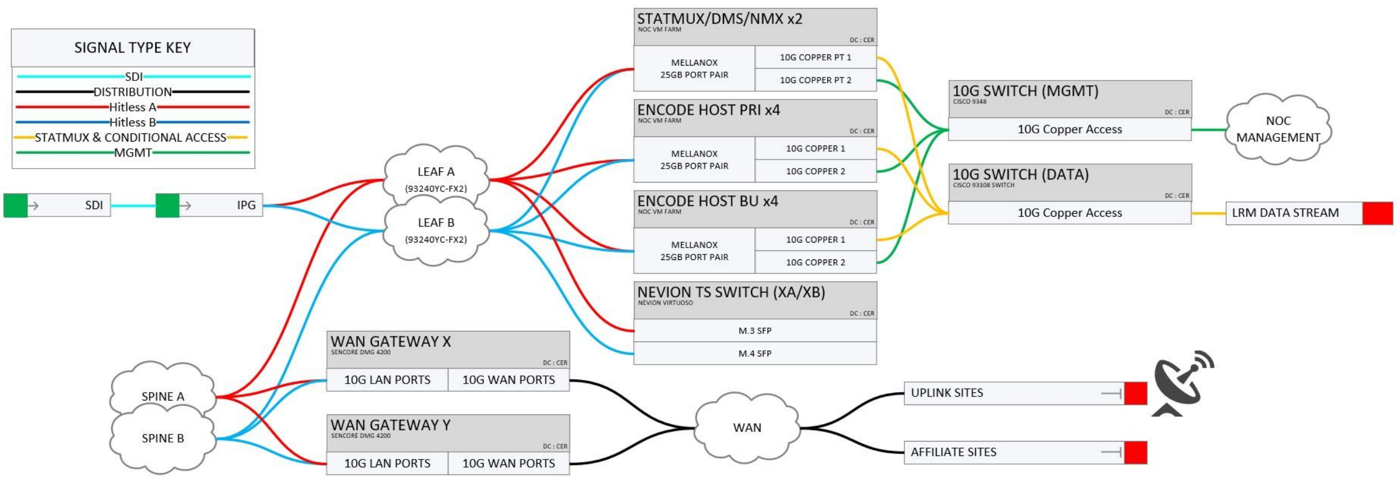


Figure 3 - Ecosystem Connectivity View

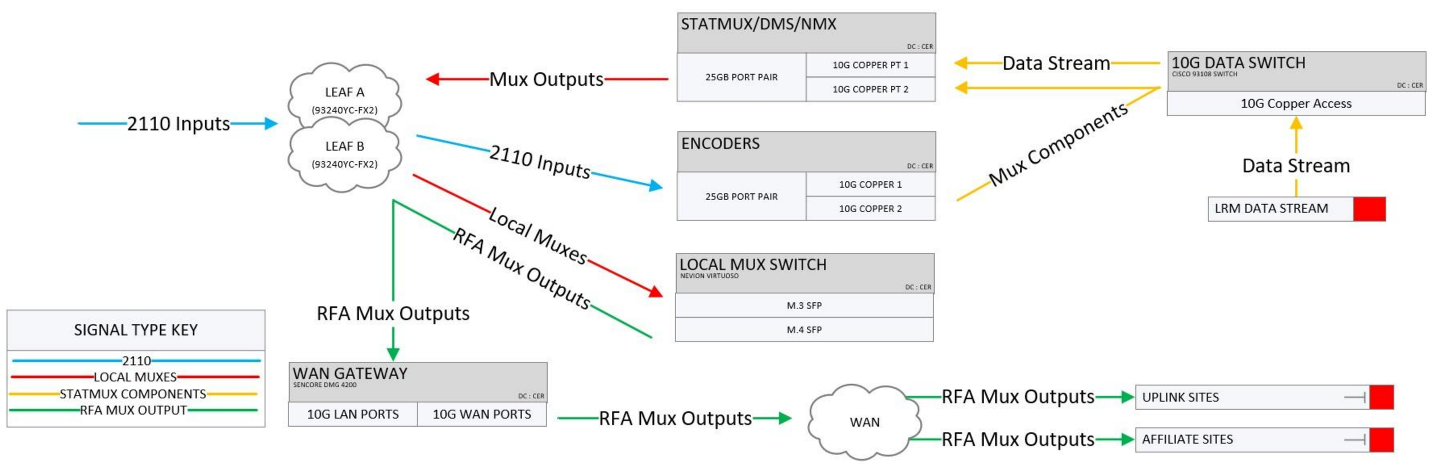


Figure 4 - Simplified Signal Flow View

Generic Header about the compression details

The Harmonic compression platform for Skypath exists entirely as software. That software currently runs on a cluster of Cisco C220M5 servers. These bare metal servers are utilized in two distinct ways; the XOS encoders run as a Kubernetes cluster, while the Prostream, DMS, and NMX components are virtual machines running on ESXi. These ESXi hosts are logically grouped inside the 'DIST' clusters in vSphere.

The rack elevation below represents the entire Skypath compression ecosystem at Dry Creek. The X side is located in Rack 1004 (Left rack in following figure), and the Y side is located in Rack 1005 (Right rack in following figure). Each of these racks have discrete power distribution, and top of rack networking. The following section of this document will cover each of these named components.

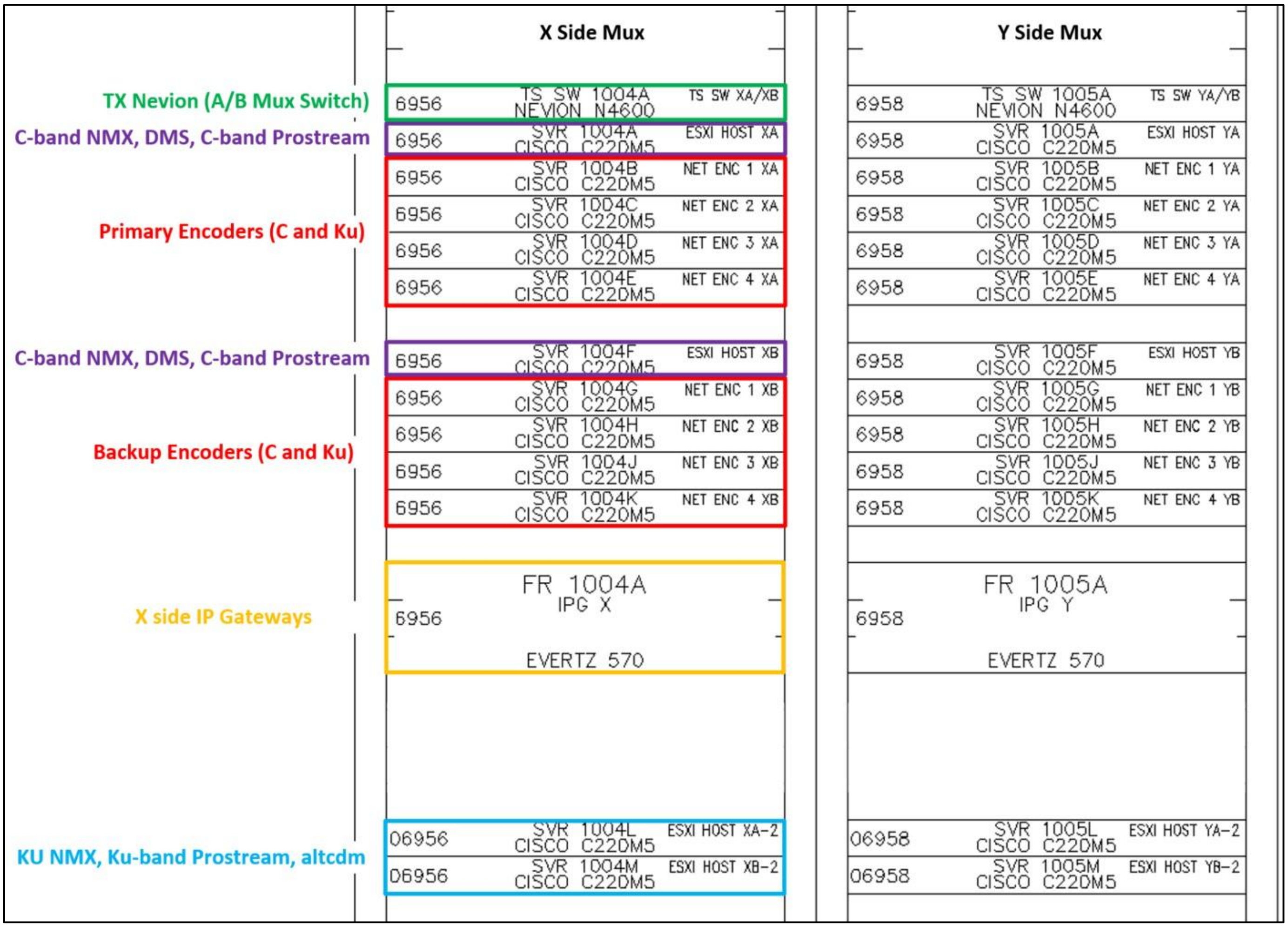


Figure 5 - Rack Components Configuration

Release Chain and 2110 Gateways (Zone #1)

The NBC channels used by Skypath are generated by Snell playout servers. These servers play file-based content like shows and advertising, integrate live video inputs like sports, and add all the ancillary data and graphics needed for the broadcast. These playout servers exist at the top of a 'chain' of components we call a Release Chain, which ultimately ends up as an input to the Skypath mux. NBC originates content using X and Y playout servers. These signals are referred to as Server X (**BLUE**) and Server Y (**RED**), and NBC employs a pair of X/Y switches to protect a failure at the playout server level. At any time, only X or Y Server signal is passed through the X/Y switches this signal is referred to as 'Program" and is illustrated in **PURPLE**.

There are occasions where NBC interrupts the playout server signals with a 3rd source called Break-In (**ORANGE**). Break-in sources are used to bypass the playout automation system entirely in the case of a failure, or to supplant the server output programming with alternative content while keeping the underlying playout intact. Both X/Y switching and Break-In process are upstream of the Mux platform and result in a discrete pair of X and Y Program signals to feed X and Y Mux encoder. This configuration allows maximum redundancy at the program creation level, while maintaining redundancy to downstream encode and uplink systems (**GREEN**).

Every NBC program in the Skypath mux has a dedicated Release Chain of identical components, allowing operators to utilize X/Y switches or break-in actions on a per-channel level. Generally, the Break-In switch point is considered the last point in which operations can maintain channel-level control of the program signals. Once the signals enter the mux ecosystem, individual program health and overall mux redundancy is managed by the Harmonic NMX system.

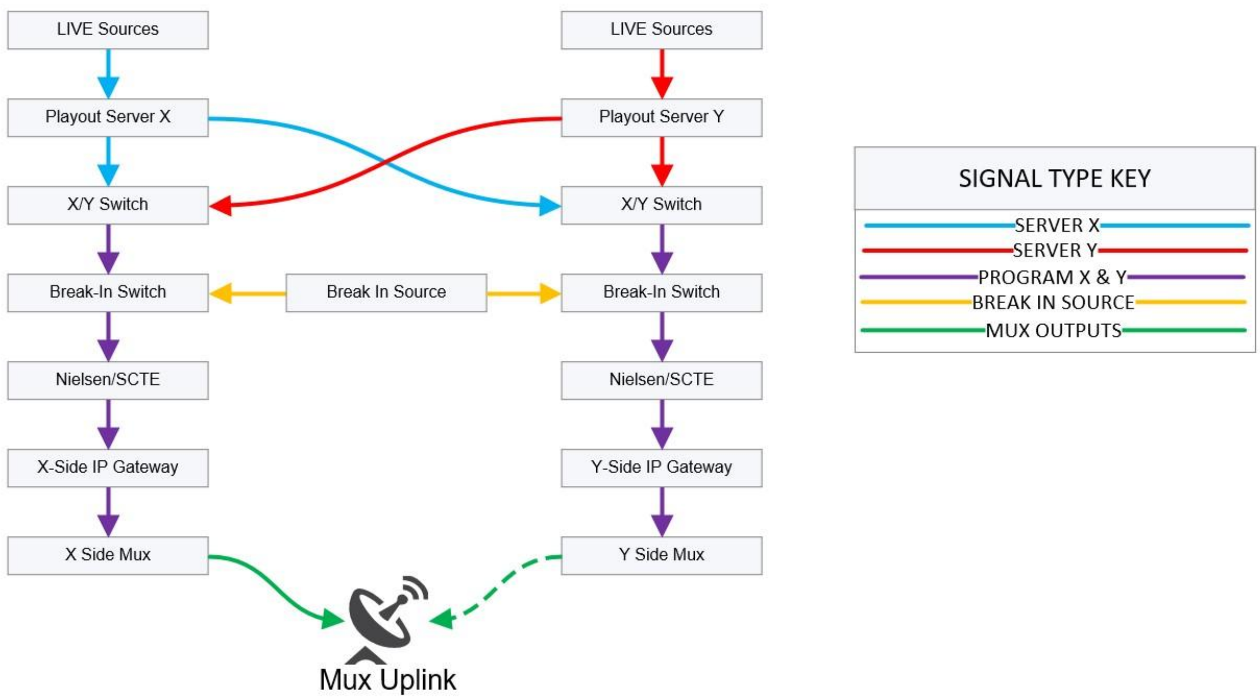


Figure 6 - X and Y Path Release Chains to Multiplexed Uplink

IP gateways dedicated to the mux inputs are located in Dry Creek. these IP gateways support multiple NBC and Telemundo channels, and the resulting ST-2110 outputs are used by multiple encoders supporting both KU and C-band muxes. The IPGs used for this are EVERTZ 570IP-X19-25G, which interface to the IPFM with 25Gb connections. The SMPTE 2110-30 audio configuration on those IPGs is a 4x4 configuration (x4 stream x4 channel). This is done to accommodate Harmonic encoder audio input best practices. Dry Creek Program outputs also hit the Evertz EXE routers via separate IPG paths, as shown in the following tables.

Note: If troubleshooting please make sure to understand the difference. See *IP Address Management* (IPAM) and Evertz *Software Defined Video Networking* (SDVN) for multicast addressing information.

IPG 1004A3	100.125.75.43

IPG 1005A3	100.125.75.44

Interface #	Service
Port 1	CH 61X (NBC E)
Port 2	CH 62X (NBC C)
Port 3	CH 63X (NBC M)
Port 4	CH 64X (NBC W)
Port 5	CH 65X (NBC 13)
Port 6	CH 66X (NBC 14)
Port 7	CH 67X (Breaking)
Port 8	CH 41X (TLMD E)
Port 9	CH 42X (TLMD M)
Port 10	CH 43X (TLMD W)
Port 11	CH 44X (TLMD OCC)
Port 12	CH 28X (COZI)
Port 13	CH 34X (TELEXITOS)
Port 14-18	Un-Used

Interface #	Service
Port 1	CH 61Y (NBC E)
Port 2	CH 62Y (NBC C)
Port 3	CH 63Y (NBC M)
Port 4	CH 64Y (NBC W)
Port 5	CH 65Y (NBC 13)
Port 6	CH 66Y (NBC 14)
Port 7	CH 67Y (Breaking)
Port 8	CH 41Y (TLMD E)
Port 9	CH 42Y (TLMD M)
Port 10	CH 43Y (TLMD W)
Port 11	CH 44Y (TLMD OCC)
Port 12	CH 28Y (COZI)
Port 13	CH 34Y (TELEXITOS)
Port 14-18	Un-Used

NMX and Encoders (Zone #2)

After a program signal from the Release Chain enters the mux ecosystem, it is encoded using statistical multiplexing under the control of the Harmonic NMX system. This is an Engineering domain, and Operations has no control over redundancy at this level.

NMX - Skypath employs *Harmonic Network Management System* (NMX) to assist in managing network distribution, employing a Network Group (NG) comprising a group of hardware devices that work together to produce one or more digital multiplexed transport streams. The NMX system is the main operational entry point for user configuration and provides the alarming structure for automatic failover between redundant components. NMX employs the following functional structure:

- **Network Group**: Nested within an NMX Site, a Network Group defines the hardware components used to execute a service plan. There are separate Network Groups for the X and Y compression systems, which operate completely independently.
- **Service Plan**: Defines the inputs, processing, and outputs of the content flowing through Network Group devices. Also called a service 'map' because of the clear flow of signal through the components. In the Figure 7 below, program inputs are shown flowing in from the left, passing through encoders, through a network switch, into the ProStream units and out through output switches producing an XA and XB mux.

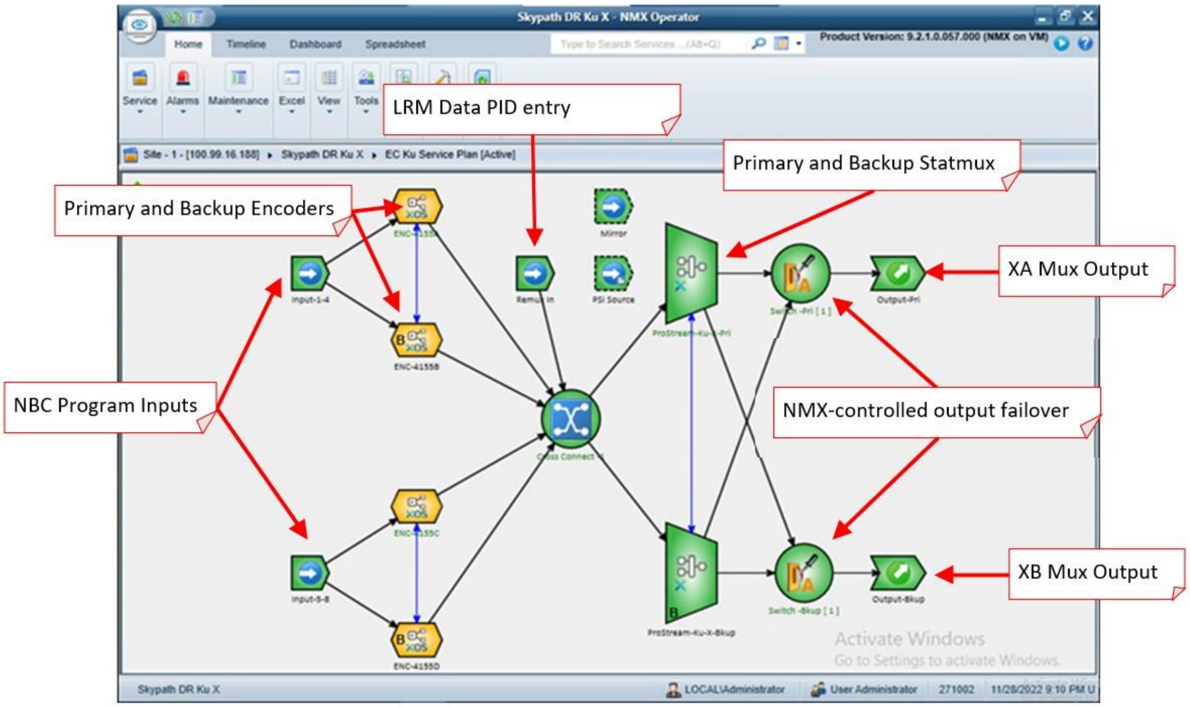


Figure 7 – Skypath Service Plan/Map

Component Redundancy in NMX Designer

Functional component redundancy is managed by NMX, with 1+1 TS streams coming from two different paths, handling different delays, or coming from two different (equipment) sources.

To establish redundance in NMX:

1. In NMX Designer, on the Network Map, draw a line between two Harmonic devices that you want to operate in a redundant pair. This is the blue line shown below.

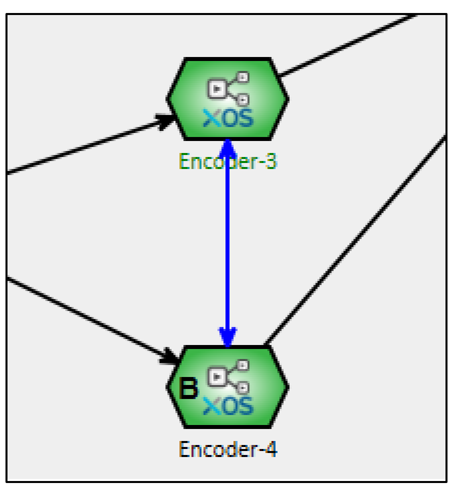


Figure 8 - Defining Redundant Pairs

2. Select the top Harmonic device.
3. In the *Properties* pane, *General Properties* section, set *Is Backup* to *False* (primary)
4. Select the bottom Harmonic device, and in the *properties* pane, *General Properties* section, set *Is Backup* to *True* (backup)

External stream inputs

In addition to managing encoders and mux outputs, the Skypath NMX system also maps the LRM scheduling data stream input. The LRM Data Carousel (aka *sctemc*) is configured to output a data stream containing local affiliate scheduling data. This stream is integrated into the mux as a *Remux In* that connects to the virtual cross connect (see Figure 9 below) and is therefore made available to the ProStreams to be included in the final mux output. Remux inputs are generally fixed-bitrate flows that do not participate in video statmux pools. More on the LRM data PID in the Skypath Document *LRM and Scheduling Guide*.

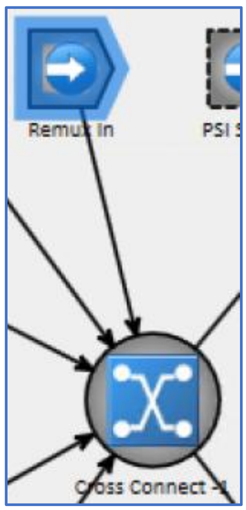


Figure 9 - Remux In and Virtual Cross Connect

Common NMX Tasks

Use the following IP addresses in the following table to access Skypath NMX resources.

NMX Virtual IP Addresses - Skypath Production at Dry Creek (X and Y paths)	
SES 3/21 X	100.125.75.55
SES 3/21 Y	100.125.75.56
SES 11/22 X	100.125.75.57
SES 11/22 Y	100.125.75.58
Skypath Disaster Recovery (DR) IP Addresses at Englewood Cliffs	
SES 3/21 X	100.99.16.189
SES 3/21 Y	100.99.18.189

SES 11/22 X	100.99.16.181
SES 11/22 Y	100.99.18.181

To access NMX:

1. RDP into dejumpboxes.tfayd.com.
2. RDP into a VM behind the site's (EC or DC) firewall.
3. RDP into the NMX
4. Launch *NMX Operator*.

Note: When finished, please log out of the NMX's RDP or VNC session. Also, be aware that NMX does not need an active Windows login session to operate.

Credentials - Credentials for RDP or VNC connections to the NMX VM and NMX Operator should be in your department's credential store (Bitwarden or KeePass).

Executing Primary Tasks on NMX Operator

The four primary tasks you will perform with NMX Operator include:

- Checking Alarms and Determining Alarm Status
- Retrieving a Techdump when something goes wrong.
- Manually Failing-over a component.
- Enabling and disabling services.

Checking Alarms and Alarm Status

Any device on the Service Plan whose color is not green indicates an alarm. In the screen shot below, the Electra XOS encoders have an active alarm.

To check alarm status:

1. Logging into NMX with steps illustrated above.
2. Get information on a device's active alarm by selecting it and displaying the Current Alarms pane (Figure 10, on the following page).

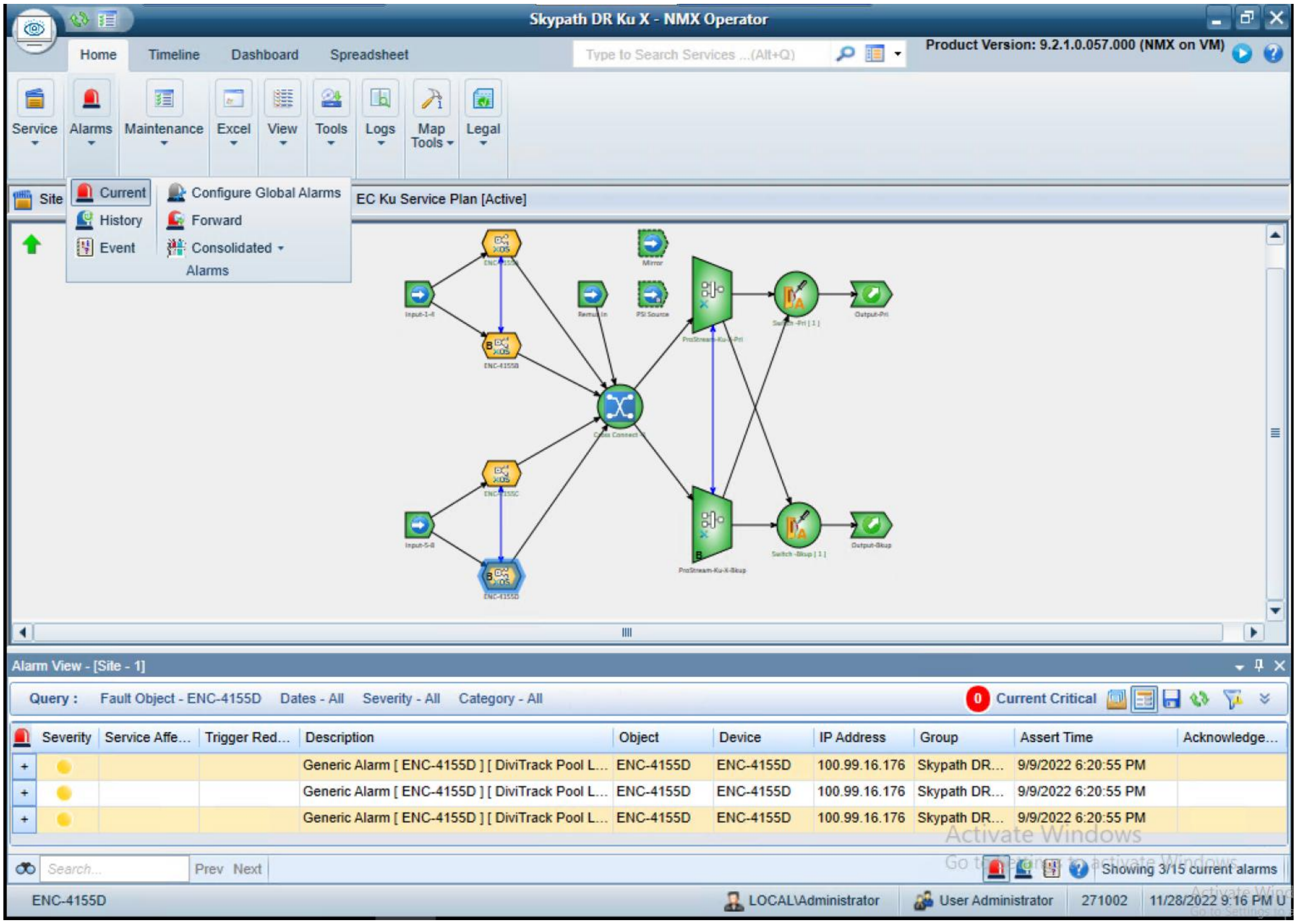


Figure 10- Displaying Current Alarms

In the screen shot above, the user selected/clicked on ENC-4155D, then went to Alarms in the ribbon at top and selected Current from the drop-down menu. The Alarm View pane shows the active alarms for ENC-4155D. If you need to see historical alarms for a device, click Alarms in the ribbon and select History. Both Current and History can be displayed at the same time.

Retrieving Tech Dumps

When a system or device error occurs, information about the error will be contained in a system-generated *Techdump* file.

Note: Techdump files are typically provided to *Harmonic Support engineering* to assist Skypath engineering with troubleshooting system issues.

To retrieve a Techdump:

1. launch and log into NMX Operator, right-click on an *Electra XOS* to display the options menu, and select *Get Device Logs*, as shown in the following figure.

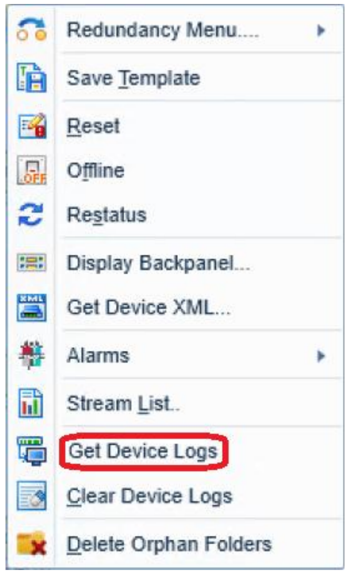


Figure 11 - Options Menu / Get Device Logs

2. When prompted, specify a location where the Techdump will be saved, then wait several minutes for the operation to complete.

3. Provide the Techdump file to Harmonic Support Engineering when they request it.

Manually Failover a Component

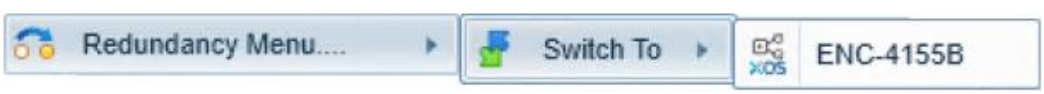
Failovers typically occur when a module requires service-related tasks, such as replacing fans or memory, or for software upgrades. Encoders, in particular, are failed over during software upgrades.

To change which device is the active device:

1. Launch and log into NMX Operator.

2. Right-click the currently active device.

3. Select *Redundancy Menu*, and *Switch To*, then select the preferred backup device, as shown in the following figure.



Additional Encoder Options Menu Functions

In addition to the manual redundancy switching:

- The *Reset* command executes a soft reboot of the encoder. **Note**: This command is service effecting and takes several minutes to complete.

- The *Restatus* command reloads the encoder provisioning information from NMX.

  Note: *Restatus* is not service-effecting. However, it will often clear a “stuck” reporting state.

Disabling and Enabling Services

Individual services or component parts of a service can be Disabled and Enabled.

To Disable or Enable a Service or Component:

1. In the Input Services section, right-click on a service to display the *Service Options* menu (Figure 12, below). Issuing a *Disable* command followed by a re-*Enable* will often clear an incorrectly reported status or restore a failed service or component stream within the multiplex. **Note**: This is a service-affecting command.

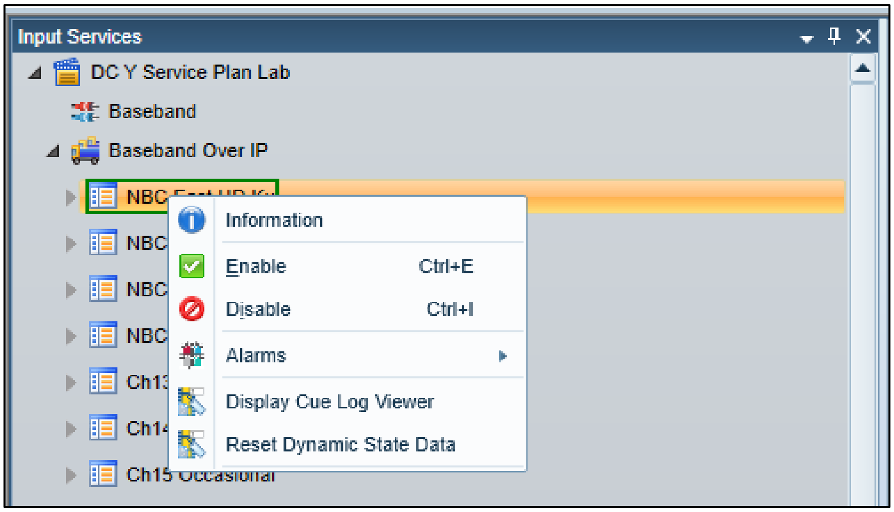


Figure 12 - Service Options Menu

Encoders - Harmonic Electra XOS processors are used by Skypath to compress and encode uncompressed video from the channel release chain. Figure 13, below, provides the NBC channel/encoding processor configuration for the X-path (duplicated for the Y path). Each server running XOS is responsible for up to 4 program encodes at a time. This multi-service approach maximizes encoder density and efficient use of available compute resources.

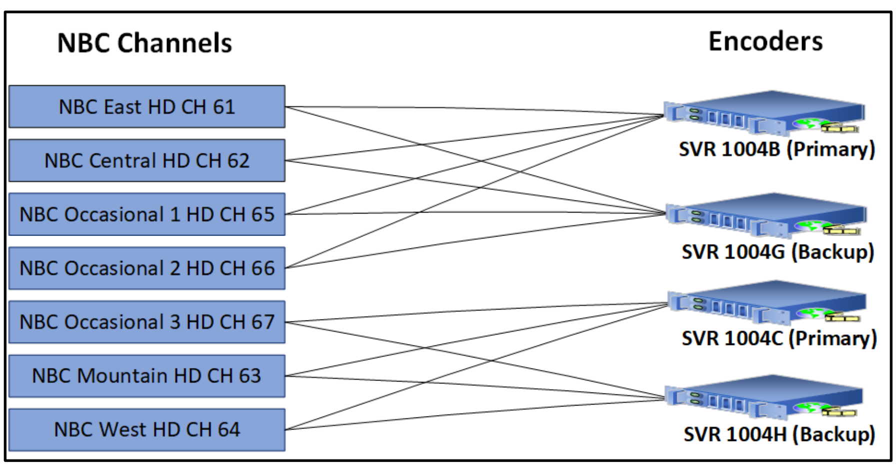


Figure 13 – Example of NBC Channel Encoding

Note: Cozi and Telex Mezz feeds are encoded in the Ku system, but not as part of the multiplex.

Each Skypath encoder is responsible for encoding multiple channels, as well as participating in the statistical multiplexing process which aims to maximize video quality with finite bandwidth. The encoder supplies complexity information to the mux, the mux evaluates the complexity information from all encoders, then sends encoding bit rate instructions back to the encoder. This process is covered in more detail inside the ProStream section of this document.

The following table provides encoder identification and addressing information, including *Cisco Integrated Management Controller* (CIMC) Addresses.

Rack Module	Name	XOS MGMT IP	CIMC	Side	Role
SVR 1004B	ENC 1 Xa	100.125.75.7	100.125.251.212	X	Primary Ku CH 61, 62, 65, 66
SVR 1004C	ENC 2 Xa	100.125.75.8	100.125.251.213	X	Primary Ku CH 63, 67, 64, 28
SVR 1004D	ENC 3 Xa	100.125.75.9	100.125.251.214	X	Primary C CH 61, 62, 41, 44, 34
SVR 1004E	ENC 4 Xa	100.125.75.10	100.125.251.215	X	Primary C CH 63, 64, 43, 42, 28
SVR 1004G	ENC 1 Xb	100.125.75.17	100.125.251.217	X	Backup Ku CH 61, 62, 65, 66
SVR 1004H	ENC 2 Xb	100.125.75.18	100.125.251.218	X	Backup Ku CH 63, 67, 64, 28
SVR 1004J	ENC 3 Xb	100.125.75.19	100.125.251.219	X	Backup C CH 61, 62, 41, 44, 34
SVR 1004K	ENC 4 Xb	100.125.75.20	100.125.251.220	X	Backup C CH 63, 64, 43, 42, 28
SVR 1005B	ENC 1 Ya	100.125.75.27	100.125.251.222	Y	Primary Ku CH 61, 62, 65, 66
SVR 1005C	ENC 2 Ya	100.125.75.28	100.125.251.223	Y	Primary Ku CH 63, 67, 64, 28
SVR 1005D	ENC 3 Ya	100.125.75.29	100.125.251.224	Y	Primary C CH 61, 62, 41, 44, 34
SVR 1005E	ENC 4 Ya	100.125.75.30	100.125.251.225	Y	Primary C CH 63, 64, 43, 42, 28
SVR 1005G	ENC 1 Yb	100.125.75.37	100.125.251.227	Y	Backup Ku CH 61, 62, 65, 66
SVR 1005H	ENC 2 Yb	100.125.75.38	100.125.251.228	Y	Backup Ku CH 63, 67, 64, 28
SVR 1005J	ENC 3 Yb	100.125.75.39	100.125.251.229	Y	Backup C CH 61, 62, 41, 44, 34
SVR 1005K	ENC 4 Yb	100.125.75.40	100.125.251.230	Y	Backup C CH 63, 64, 43, 42, 28

Basic ASIC Encoder Configuration and Integration to NMX

Once NMX detects an Electra XOS encoder IP address included in the Service Plan, it will take control of that Electra XOS encoder and push the NMX configuration.

In the event both hard drives for an Electra XOS were to fail or a full host replacement was necessary, you must image an .iso file of a specific version of Electra XOS to a new RAID1 volume in the Cisco UCS C220 M5. Once that task is complete, the engineer then uses the Cisco UCS C220 M5 CIMC KVM to log into the device (check your department's credential store) and runs the following command to set the management IP address.

```
nmi set network --ip <ip-address-of-device>/<CIDR-size> --gw
<gateway> --type none --disable-bak
```

Once this command is executed, run:

```
nmi get network
```

to confirm the management network configuration.

If the Electra XOS has been removed from the Service Plan and then re-added, it is important to configure GbE 2 of the Electra XOS to *Auto*. **Note**: Failure to do this will likely result in CC errors being introduced to any active streams on the same island switch the moment NMX takes control of the new Electra XOS encoder.

1. From the Network section of NMX Designer, right-click on the *Electra XOS*.
2. Select *Display Backpanel*

Note: This is a representation of our Cisco hardware. *Dual GbE* is the Mellanox Connect-X5 PCI-e card used for SMTPE-2110 input. *CPC-C* represents the onboard chassis 10Gb ports, used for Management and statmux data.

3. At right, in the *Network View* section, Select *GbE 2 Rx* and set the Speed/Duplex to Auto.
4. Select Tx for GbE 2 and set the Speed/Duplex to Auto.

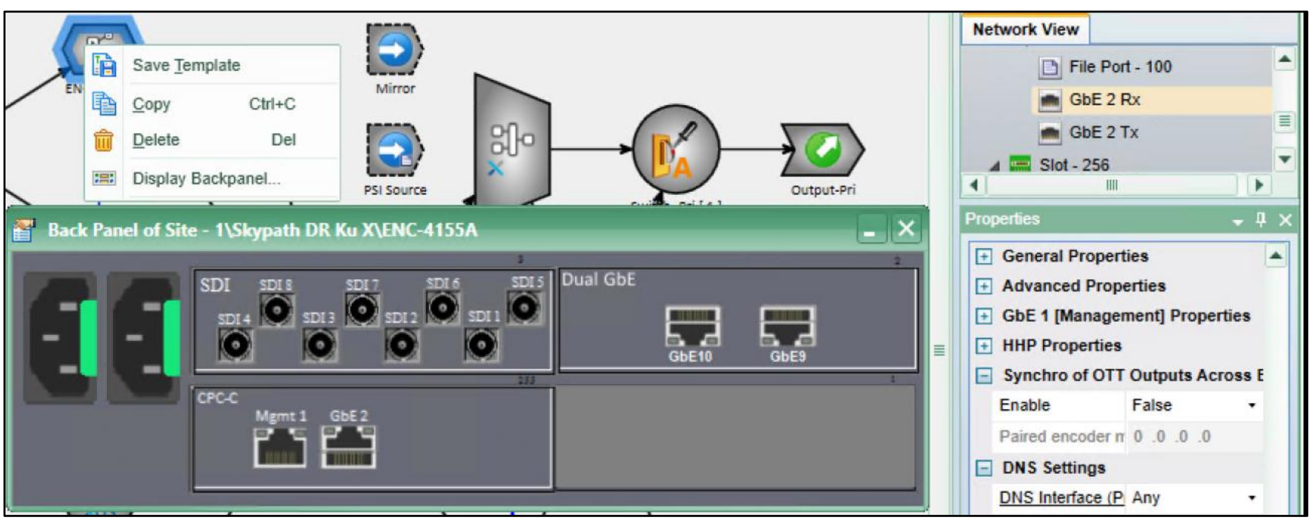


Figure 14 – NMX Designer / Configuring GbE 2 Mellanox Card

Note: The following Cisco server hardware spec. applies to both Encoders and ESXi hosts, for easy spare pool management.

Chassis:	UCSC-C220-M5SX
Processor Type:	Dual Intel Xeon Gold 6248 CPU @ 2.50GHz (40 Cores)
Memory:	192 GB (12 x 16 GB DIMM)
Storage:	x2 400 GB SSD (Single RAID 1 volume)
Networking:	Mellanox ConnectX5 2 x 10/25 GB SFP+ NIC, onboard 2 x 10 Gbps RJ45 NIC

NBCU has enabled *Reduced Latency* mode for the HD services on Electra XOS encoders. Reduced latency does not apply to (and is not available on) SD encodes. This was necessary to match the latency of the Harmonic compression platform to that of the previous compression platform. When Reduced Latency mode is enabled, the following prompt displays.

Note: As Tuesday, November 29th, 2022, NBCU is using a version of NMX (9.2.1.0.057.000) that displays a false, Generic Alarm that DiviTrack Pool Latency and Transcoding Latency Not Match. This bug is supposedly fixed in future release NMX 9.4.1.

Harmonic compression systems that use SMPTE 2110 IP baseband inputs have a combined input buffer for each service that accommodates both SMPTE 2022-7 input offsets and any offsets that may be present between the various 2110 essence streams. This buffer, called *Maximum Support Delay*, is user adjustable (see Figure 15, below).

We have also used this buffer to time the mux outputs to roughly match legacy systems. In the case of the C Band muxes, additional time has been added to this buffer to roughly time the decoded C Band output at a station to the Skypath Ku Band output as shown in Figure 15, below.

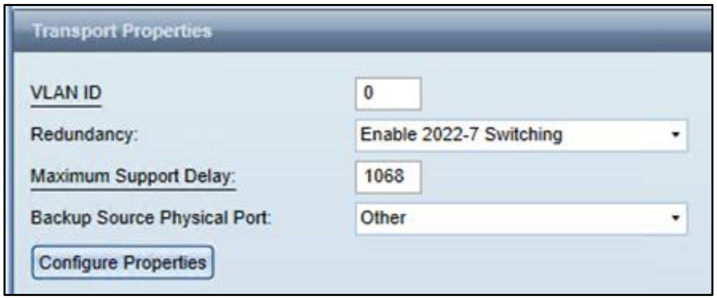


Figure 15 - Transport Properties - Maximum Support Delay

Encode Characteristics.

The following table defines the specifications and relevant parameters for Skypath HD and SD encoding, respectively.

NBC HD SES-11/22C and SES-3/21Ku

Parameter	
Video Bit Rate:	Variable from 0.5 to 25 Mbps H.264 (MPEG-4 AVC) High Profile
Resolution:	1920x1080i
Frame Per Second:	29.97
Chroma Sampling:	4:2:0 8 bit
Structure:	Variable Open GOP Mini GOP Length (M) 4 GOP Length (N) 64
Configuration:	Scene change detection will result in an I frame CABAC enabled 1 slice per picture Video PID: <service-id-number>0
Audio Encoding:	Dolby AC-3 Constant bit rate Audio 1 PID (AC-3 5.1): <service-id-number>1 (Program Audio 5.1) Audio 2 PID (AC-3 stereo): <service-id-number>2 (SAP/VDS/AD/mono) Audio 3 PID (AC-3): <service-id-number>3 (future use)
Audio Bit Rate:	Audio 1 448 Kbps Audio 2 192 Kbps Audio 3 96 Kbps
Audio Sample Rate:	48.0 kHz
Data:	Captions: 608/708 SCTE 35: PID <service-id-number>5

NBC SD services on SES-11/22C and SES-3/21Ku

Parameter	
Video Bit Rate:	Variable from 0.5 to 6 Mbps MPEG-2 - Main Profile, Main Level
Resolution:	720x480i
Frame Per Second:	29.97
Chroma Sampling:	4:2:0 8 bit

Structure:	Variable Open GOP
	Mini GOP Length (M) 3
	GOP Length (N) 15
Configuration:	Scene change detection will result in an I frame
	Custom Matrix / BVOP
	1 slice per picture
	Video PID: <service-id-number>0
Audio Encoding:	MPEG-1 Layer II
	Constant bit rate
	Audio 1 PID (MPEG-1 Layer II): <service-id-number>1 (Program Audio 2/0)
	Audio 2 PID (MPEG-1 Layer II): <service-id-number>2 (DTMF)
COZI TV SD only (SES-11/22C-Band)	Audio 3 PID (AC-3): <service-id-number>3 (SES-11/22C-Band only, encrypted for use by a single MVPD with a special Nielsen SID)
	Audio 1 192 Kbps
Audio Bit Rate:	Audio 2 96 Kbps
	Audio 3 192 Kbps
Audio Sample Rate:	48.0 kHz
Data:	Captions: 608
TeleXitos only	SCTE 35: PID <service-id-number>5
	SMPTE 2038 VANC: PID <service-id-number>6
COZI TV only	*Likely useless since it references line numbers in the HD-SDI input to the encoder

CTS Data Carousel (SES-3/21Ku-Band only) Service Channel 68

normalPriority	PID 42 (decimal), 0x2A – 150400 Bytes per second / 150.4 Kbps
highPriority	PID 53 (decimal), 0x35 – 809846 Bytes per second / 809.846 Kbps

Prostreams (Zone #3)

Statistical multiplexing (Statmux) is a process used in Multi Program Transport Streams that allocates transport stream bandwidth between more complex and less complex encoding content. Less complex content sacrifices bandwidth in order to provide it for more complex content. The more complex content benefits from the larger allocation of bandwidth. And the total size of the multiplex is kept within the parameters needed for satellite transmission. Harmonic calls this process *DiviTrack*.

Multi-pass encoders that are typically used for content distribution will use an early encoding pass to provide complexity information to the multiplexer. The multiplexer uses the complexity data for each service and references this to encoding pool rules that might lend different desired quality weights to service channels.

Control data is then sent back to the encoder to tell it what bit rate should be used to encode a service at any given moment. Note that this process applies only to video content. All of the MPEG table information, descriptors, ad insertion data, and most importantly, the audio streams are <u>not</u> subject to the stat mux process as they have fixed bit rate allocations.

Note: NMX provides *logical* switching to indicate which Prostream X mux is providing the (statmux) commands, but the TX (transmit) Nevion switch performs the *actual* switching of the stream, not the

NMX switch. Figure 16 and Figure 17 below illustrate the signal process flow from input to uplink, as viewed in the native Nevion switch graphical interface.

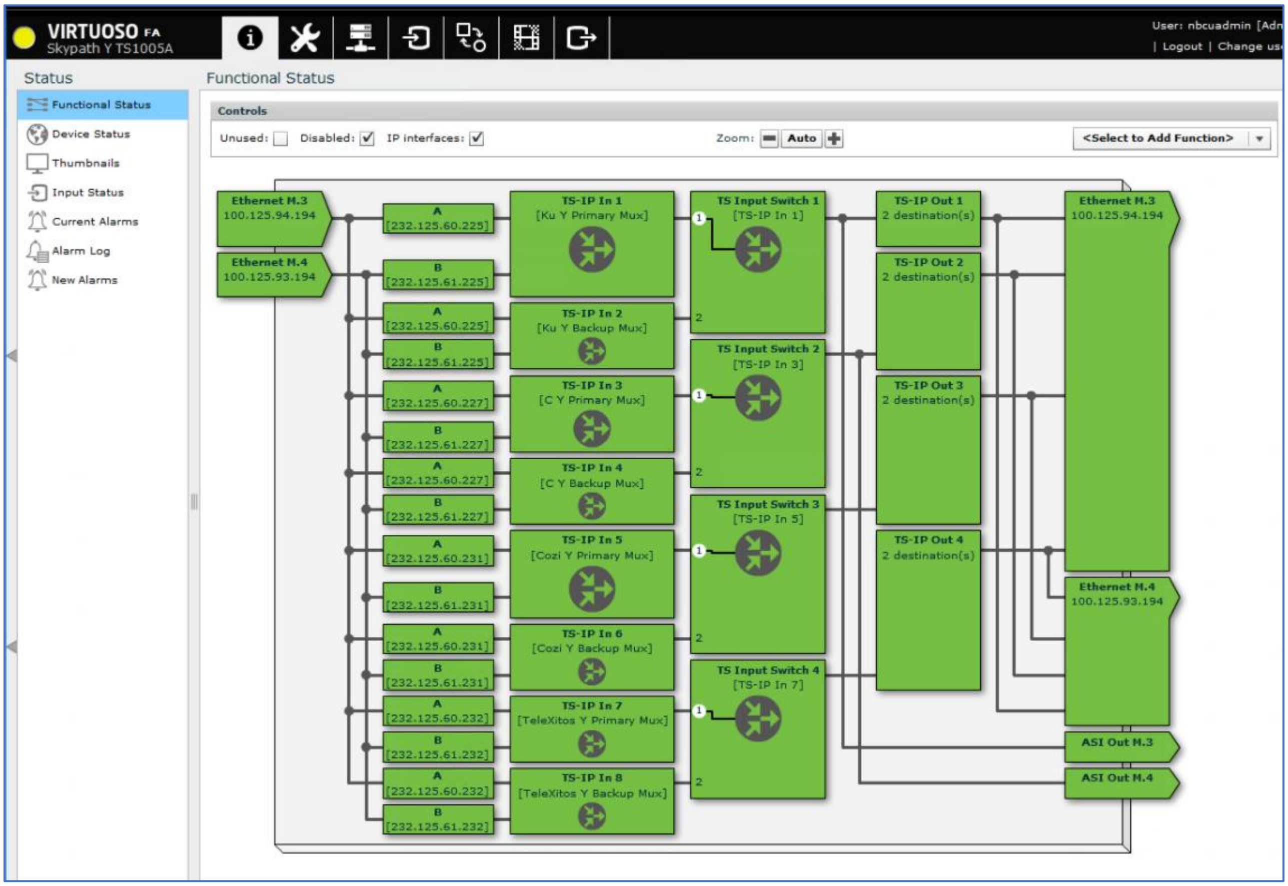


Figure 16 - NMX Inputs for Uplink

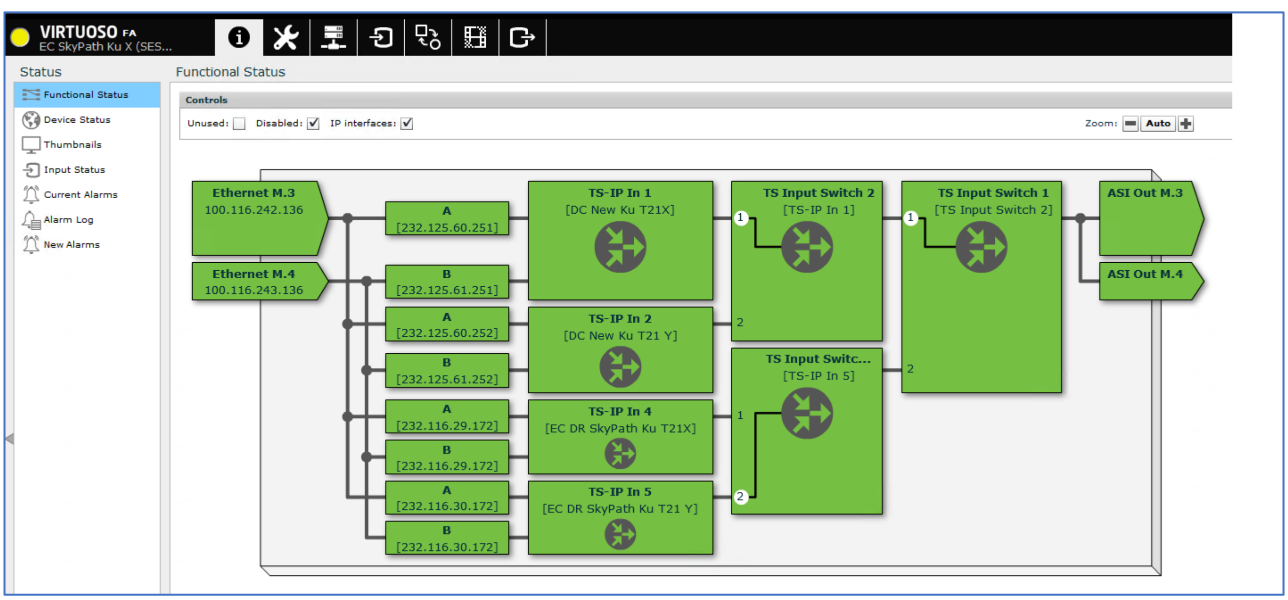


Figure 17 - NMX/Nevion Switch Transmit for Uplink

Stat Mux Pools

Encoding changes may require a stat mux pool to be removed, then rebuilt after the encoding parameter change. Under most circumstances however we would not need a new pool. It would need adjustment if any of the fixed bit rate services changed (an added audio service for example).

Stat mux pool size is determined by the simple equation:

Total multiplex size – fixed bit rate allocations = stat mux pool size.

The following table provides Service Channel minimum/maximum bit rates and Pool Priority.

Service Channel Name	Service Channel Number	Minimum Video Bitrate (Mbps)	Maximum Video Bitrate (Mbps)	Pool Priority
Cozi TV Ku	28	0.5	6	High
NBC East HD	61	0.5	25	Medium
NBC Central HD	62	0.5	25	Medium
NBC Mountain HD	63	0.5	25	Medium
NBC West HD	64	0.5	25	Medium
NBC Occasional 1 HD	65	0.5	25	Medium
NBC Occasional 2 HD	66	0.5	25	Medium
NBC Occasional 3 HD	67	0.5	25	Medium
CTS Data Carousel	68	0.1504	1	Not Applicable

To create or modify a StatMUX Pool:

1. In NMX Designer, select the *Service* tab in the ribbon (Figure 18, below).

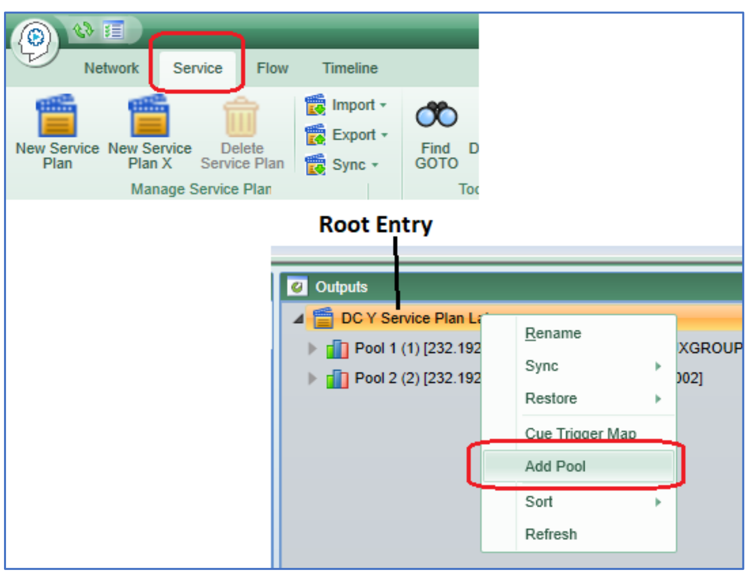


Figure 18 - Adding a Pool

2. In the *Outputs* pane (refer to figure above) Right-click on the root entry, (highlighted in yellow when selected).
3. Select *Add Pool* (or select another existing pool).
4. Right-click on the Pool you just created/selected and select *Add Video.*
5. In the *Select Video Stream(s)* window, select the streams to add to the Pool and click OK.

Overall mux bitrate is determined by the RF modulation scheme used for uplink and downlink, currently that data rate for each mux is roughly 72 Mbps.

MUX output failover and Stub Switches

The stub switches are virtual 2x1 switches that indicate which PSX is the active DiviTrack rate controller. Both StatMUXes are always active (a *hot/hot* configuration). The CER Nevions have TS inputs with SIPS configured to consume a StatMUX's primary output (flow A) and a StatMUX's mirror output (flow B). The CER Nevions select which StatMUX is being output to the uplink Nevion.

Distribution Management System (DMS) & IRDs (Zone #4)

DMS lives on a pair of redundant servers in the Dry Creek compression clusters (ESXI). To avoid additional technical complexity, there is no DMS on the Englewood Cliffs DR system. **Important**: If NBC is *DR Hot* with the Englewood Cliffs mux, no new IRDs can be added, authorized, or configured.

Using DMS

To use DMS:

1. Log into the DMS using the RDP. Choose an IP address for the server.

   X: 100.125.75.5

   Y: 100.125.75.25
2. After logging in, double click the DMS client icon on the desktop:
3. Reach out to **Affiliate Ops** for the username and password to use at the DMS login box.
4. After logging in, double click the database icon (red cylinder), and the *Device Browser* will display, as shown in Figure 19, on the following page.

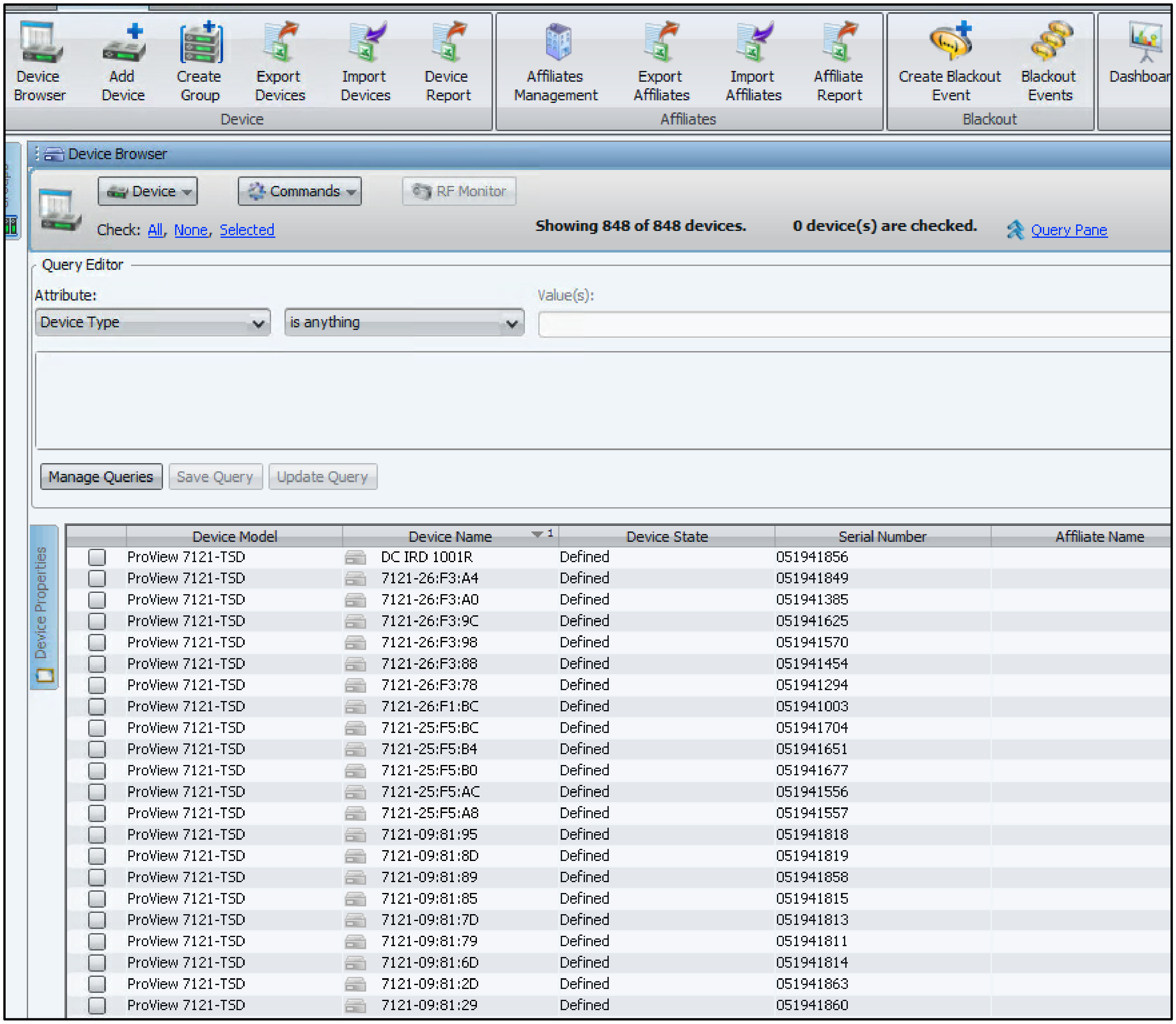


Figure 19 – Device Browser

Note: Device details can be edited in the *Device Browser*. *Device Name* and *Device State* are populated for each Skypath affiliate.

BISS encryption

NBC uses BISS encryption on both the C band and Ku band muxes. With the exception of the Cozi program and the LRM data PID, all video programs are encrypted using a shared BISS key. While BISS is not as secure as other conditional access systems, NBC's intent for utilizing BISS encryption was not to protect the on-air content but to simply force affiliates or other content consumers to contact us for and IRD refresh. By using simple BISS, with keys distributed in-band via DMS, we are taking a more active approach to controlling the IRD fleet and maintaining an accurate database regarding where our signals are being used.

A new BISS key can be pushed from DMS at any time, but the underlying rationale behind encrypting NBC may not necessitate any changes to BISS going forward.

To authorize an IRD, and confirm the BISS keys:

IMPORTANT! The following steps are provided only for confirming that the BISS Keys are generated. *Changing any of the BISS key settings may result in taking affiliate programming off the air*.

1. Log into DMS and navigate to the authorization tab on the *Device Configuration* dialog, shown in Figure 20, below.
2. In the *Input Profile* section, select the *Authorization* tab from the *Select Input Profile* drop-down menu choose the appropriate NBC profile, and at the bottom of the dialog, click *OK*.

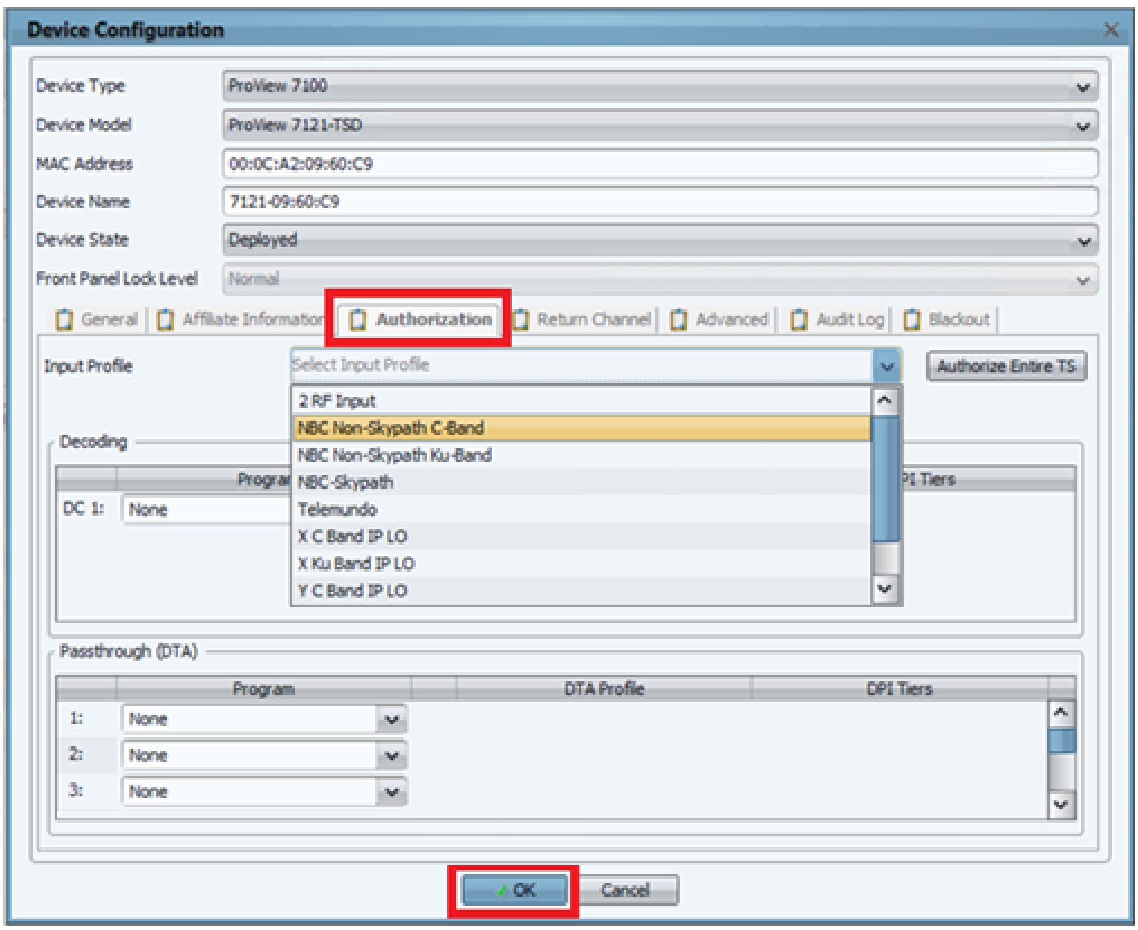


Figure 20 - Device Configuration Page

3. At the *Authorize Entire TS* dialog, button and at the bottom-center of the dialog choose *OK* to display the *Authorize Entire TS* dialog (Figure 21, below).

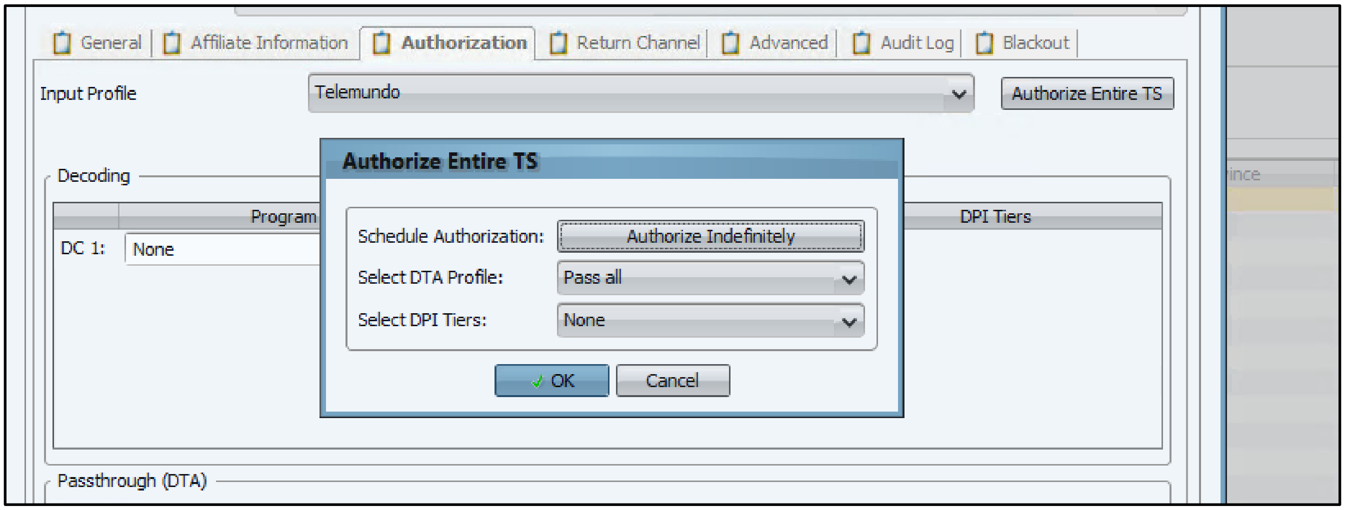


Figure 21 - Authorize Entire TS Dialog

Once the Proview IRD has been successfully authorized, it will have received 1 or 2 BISS keys from the Digital Management System (DMS). These are numbered 1 and 10 and have the *BISS-E Injected Mode* type. **Note**: Most receiving partners we will use Key 1.

4. From the DMS main page, click on *CA & BISS* Tab (Figure 22, below) from the IRD and select *BISS* to display the keys (Figure 23, on the next page).

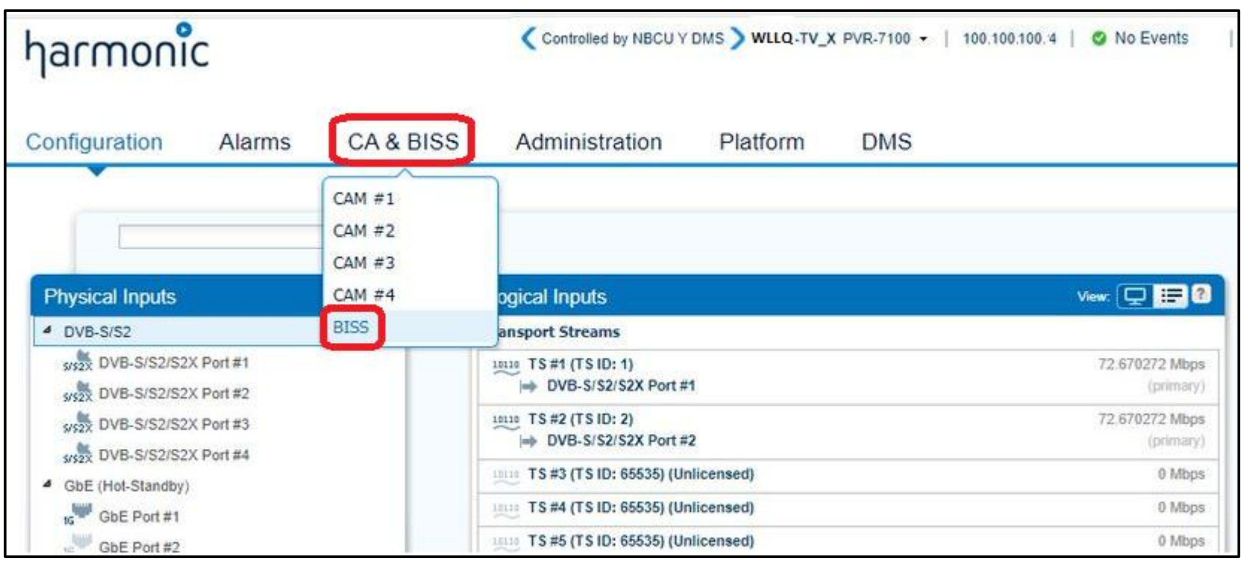


Figure 22 - Selecting BISS to View Keys

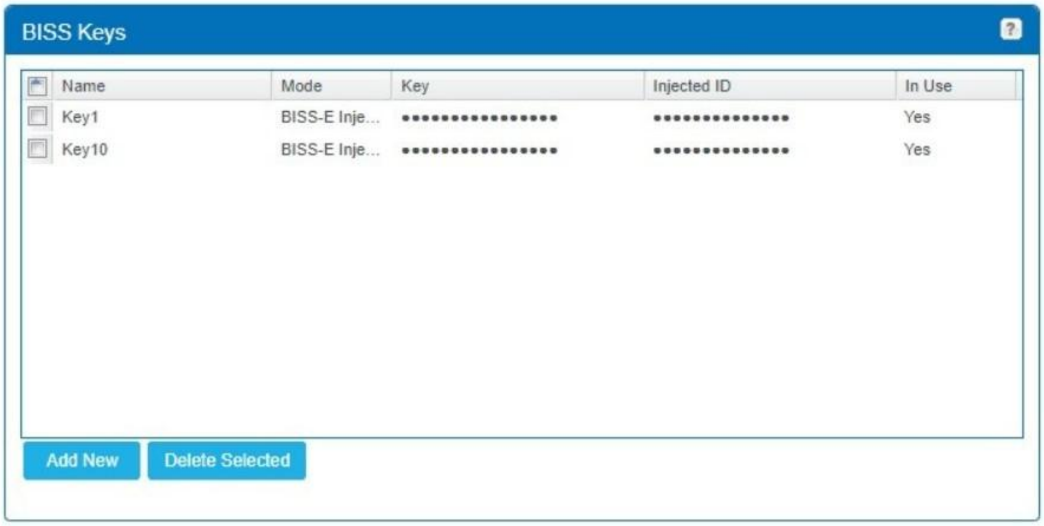


Figure 23 - BISS Keys Displayed

DMS – IRD Status

Three types of IRD status are provided by DMS, including:

- **Stand Alone** - The IRD is unaware of any DMS control info on the input.
- **Connected to** - An IRD sees the DMS control streams but is not authorized in the database.
- **Controlled by NBCU (X or Y) DMS** - This IRD addressed in the DMS database.

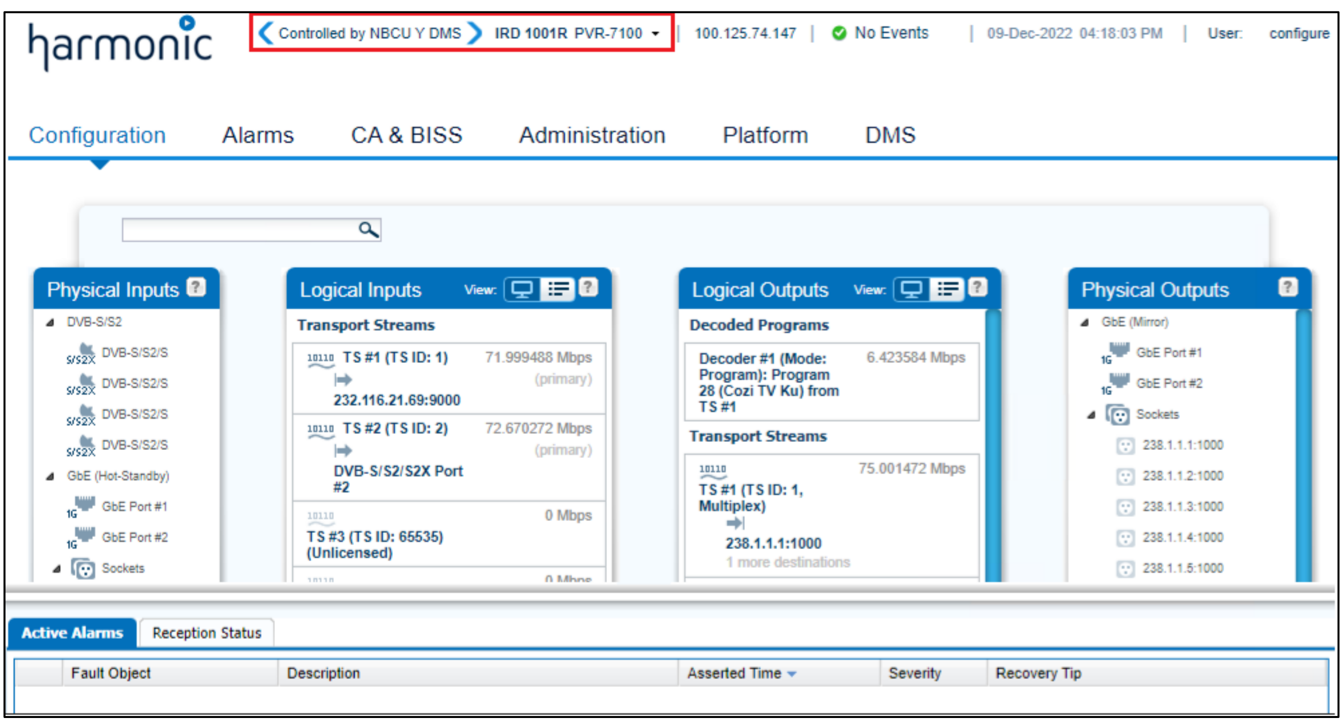


Figure 24 - IRD Status

Key Usage View - The Key Usage view, shown below in Figure 25, shows the active key with a Used By list showing the channels that are decrypted by the key. This confirms that the IRD decode path as well as the MPEGTS and ASI output paths are fully decrypted. IRDs that display 'Controlled' DMS connection status are correctly configured. An IRD that shows 'Connected' or 'Standalone' is not fully authorized in DMS.

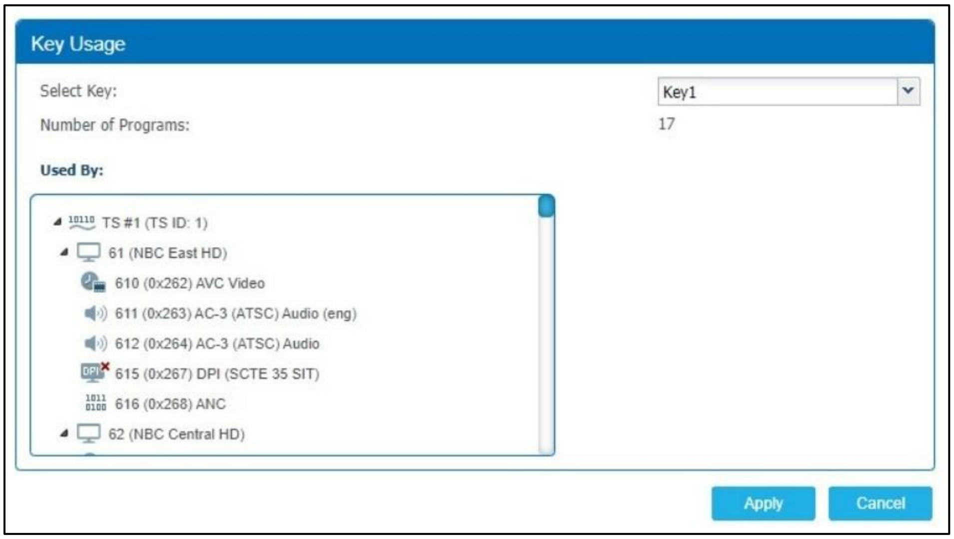


Figure 25 - DMS Key Usage View

Figure 26 – DMS Control Status

Skypath Integrated Receiver Decoders (IRDs)

The Harmonic IRDs deployed for Skypath are critical for two simultaneous workflows. These units can receive both the primary NBC Ku band mux, as well as the backup C band mux. These separate RF inputs are used in parallel and offer two distinct use cases to the affiliate (Figure 27, below).

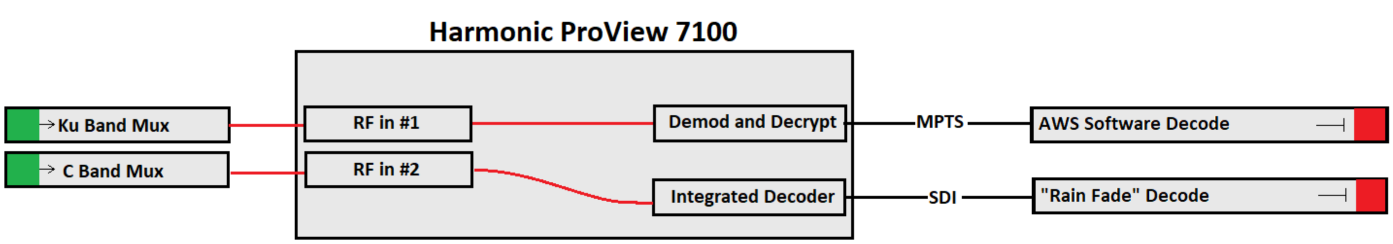


Figure 27 - ProView 7100 Dual Path Workflow

Note: A more detailed document, The *IRD & Downlink DMS Guide* covering in-depth IRD/DMS topics is available on SharePoint here.

The RF inputs are found on the DVB-S/S2 front-end module in the top left corner of the rear panel (detailed in red, in Figure 28, below).

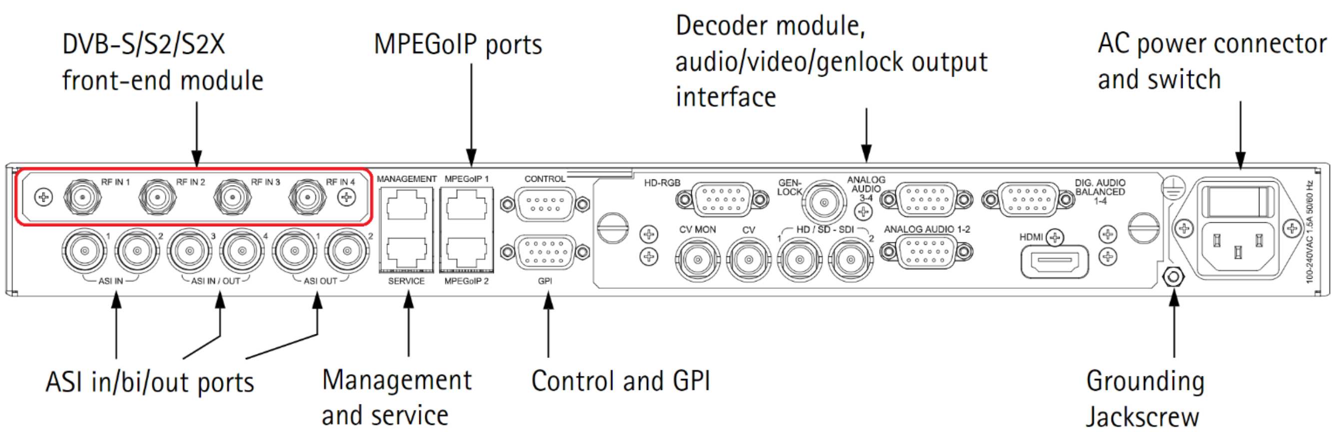


Figure 28 - Harmonic Proview 7100 Rear Panel

Typical NBC IRD configuration requires the following RF input configuration, as illustrated in the screen shots below.

RF1: Ku-Band from SES 3 Transponder 21H (12.120 GHz) L-Band 1.37 GHz
RF2: C-Band from SES 11 Transponder 22H (4.14 GHz) L-Band 1.01 GHz

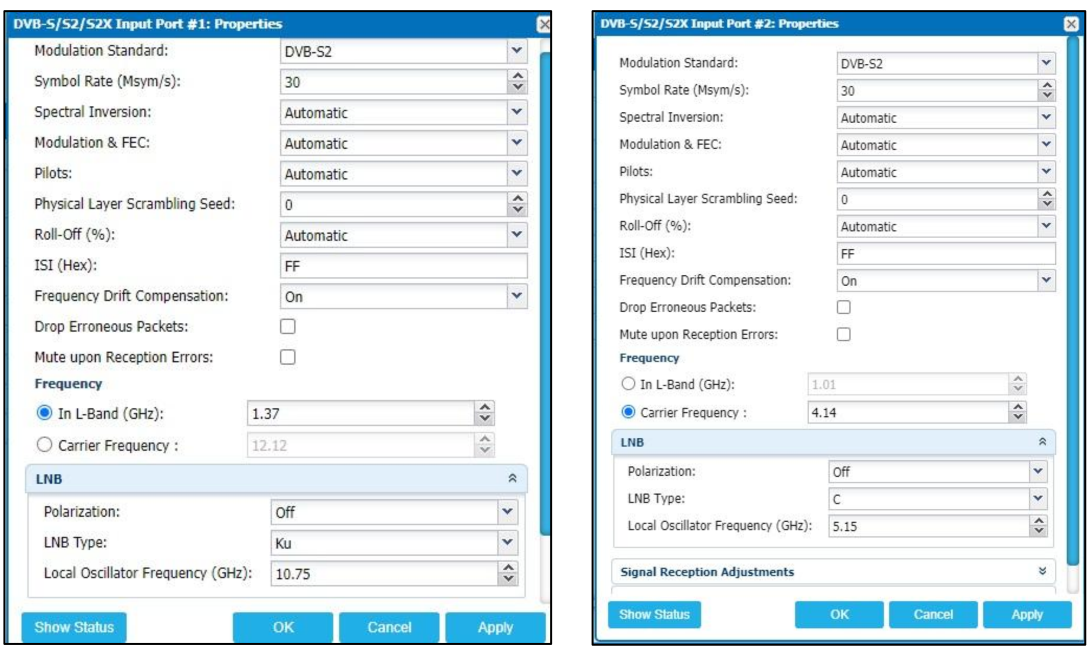


Figure 29 - Input Port Properties for Ku and C Band

Use case 1: The primary Skypath output for each affiliate is generated by a downstream Elemental Live software decoder. The primary role of the Harmonic Proview IRD for this workflow is to pull the Ku signal down from the station satellite antenna, demodulate and decrypt the signal before handing an un-encrypted MPTS stream to the software decode ecosystem downstream.

All IRDs deployed for Skypath use have the same multicast output configuration, allowing a simplified configuration of the downstream ecosystem. IP inputs and outputs are called 'sockets' in the IRD (see Figure 30, on the following page). The downstream encoders are currently configured for source specific (SSM) addressing, and our deployment scripts are responsible for matching each local IRD data port IP addresses to these generic multicast addresses.

X side IRD, KU multicast output: 232.125.18.1:9000

X side IRD, C-band multicast output: 232.125.18.3:9000

Y side IRD, KU multicast output: 232.125.18.2:9000

Y side IRD, C-band multicast output: 232.125.18.4:9000

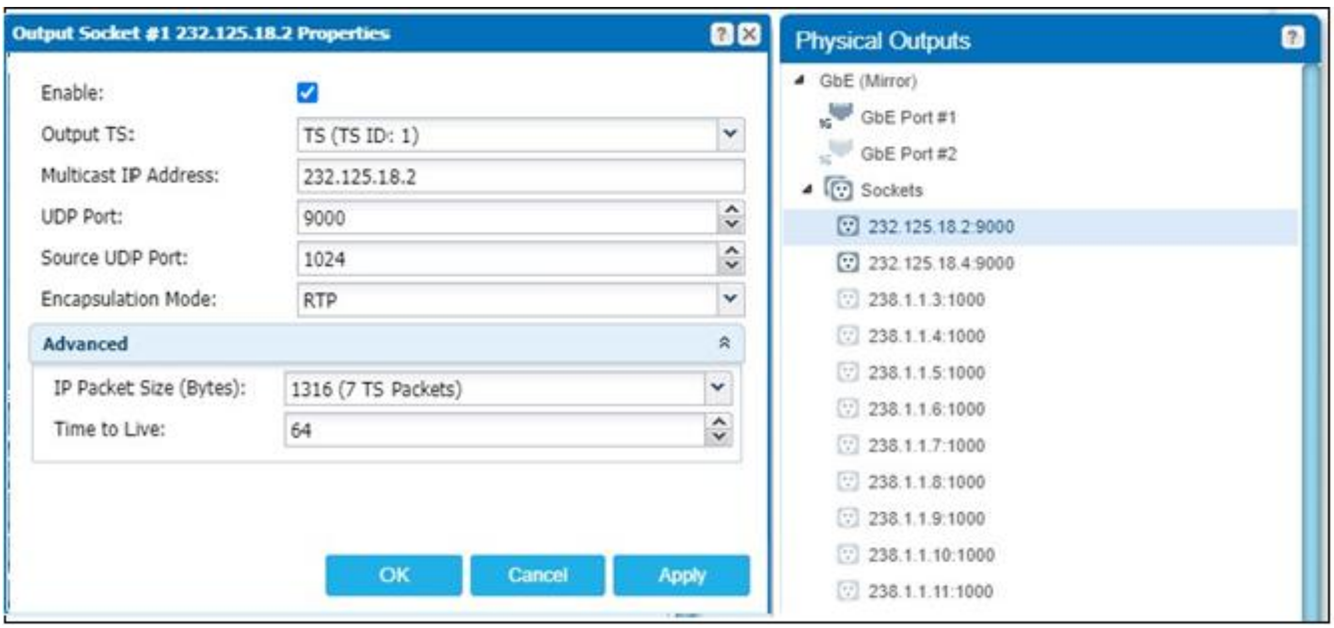


Figure 30 - Output Sockets

At the bottom of the Configuration page on the IRD there are 2 tabs. One is labelled Active Alarms and the other is labelled Reception Status (Figure 31, below). By clicking on the 'Reception Status' tab you will be presented with signal reception statistics that will provide insight as to whether any received signal is of sufficient quality for use.

When configuring the RF input ports this tab should be selected and verified before proceeding. Please refer to the Eb/No value especially (detailed in Figure 31, below).

Active Alarms | Reception Status

Port	RF Status	C/N (dBc)	Eb/No (dB)	Link Margin (dB)	BER	PER
DVB-S/S2 #1	Locked	12.99	10.11	6.37	N/A	0
DVB-S/S2 #2	Locked	12.34	9.47	5.72	N/A	0

Figure 31 - Reception Status

The critical data variables include:

Port: indicates which RF input port the data in that row is referring to.

RF Status: This should show as *Locked.*

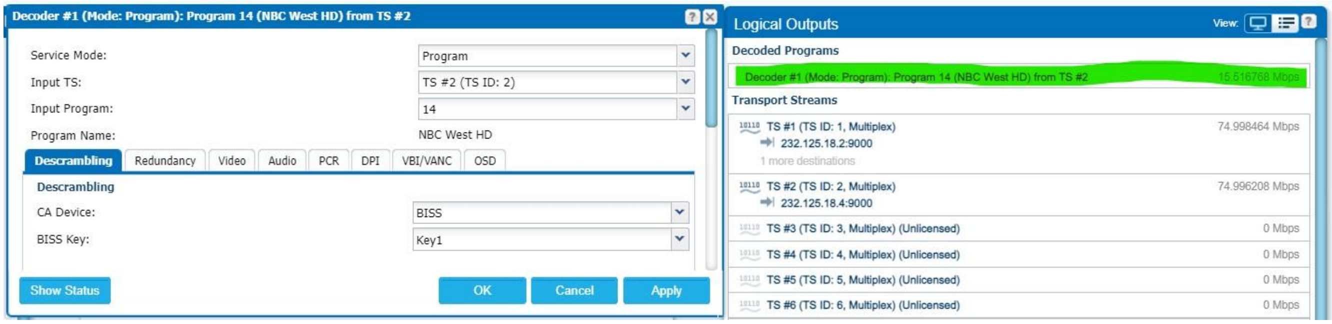
Eb/No: This value must be at least 8dB for stable delivery. Values less than 8dB will be more susceptible to weather related impact. Values less than 6dB should be considered unusable and IRD output can deteriorate to the point of failure. Station engineers should be alerted if the Eb/No value is too low.

Link Margin: Value should be at least 3dB. If any less, this should be alerted to the station engineering team to address.

PER: Packet Error Rate is the number of incorrectly received data packets divided by the total number of received packets. 0E-7 is the expected value.

So long as the RF Status indicates *Locked* and all values are highlighted in green, the signal should be considered acceptable for reception and decode. If Eb/No and/or Link Margin fall below expected levels as detailed above, this should still be referred to the station engineering team to address.

Use case 2: Ku satellite signals are prone to interruptions due to local weather. Heavy rain or snow can cause signal degradation on Ku signals due to their shorter wavelengths. This reduction in signal is generally known as 'rain fade'. To mitigate this issue, NBC provides a sub-set of our programs on a C band backup path. These C band signals penetrate rain and snow better than Ku, and as such affiliates maintain them as a rain fade backup source. The Proview IRDs take in the C band RF signal and uses the integrated decoder to provide a backup SDI source to each affiliate. This path protects both Ku weather interference as well as any downstream software-decoder issues that might occur. In a trouble situation, stations are directed to switch to this C band output while we trouble-shoot the larger ecosystem.



Many of the integrated decoder settings are pre-defined by the DMS configuration, but local stations do have the ability to change the program from the front panel of the IRD. Above is a screen shot of the Decoder configuration menu, and where it can be located on the Logical Outputs list. Descrambling using BISS key is required to get a picture, while most Video and Audio settings are left to 'automatic' mode to accommodate a front panel program change. See the IRD SOP for more detailed trouble-shooting steps for the Harmonic ProView IRDs.

Uplink Sites (Zone #5)

Overview and Strategy

Skypath Satellite Transmission, or *Uplink* is facilitated at one of 3 locations, known as Earth Stations. Only one Earth Station actively transmits to the satellite at a time, to prevent Satellite overload or damage – called *Dual Illumination* - and is carefully avoided. Skypath's three Uplinks include:

- **Titan** in Colorado, is located 7 miles from the Dry Creek Network Operations C. The Titan facility is owned and managed by Comcast.
- **UCBC** (Universal City Broadcast Center) is on the Universal Studio Lot in California. This Uplink, also known as *Golden Monkey,* is owned by NBCU but maintained by USSI at our direction.
- **Englewood Cliffs** (New Jersey) which is owned and supported by NBCU Engineering. We use multiple locations to ensure reliability, perform routine maintenance, and accommodate projects at the Earth Stations safely.

Figure 32, on the following page, illustrates the *Network Distribution Control* Overview as viewed in DataMiner.

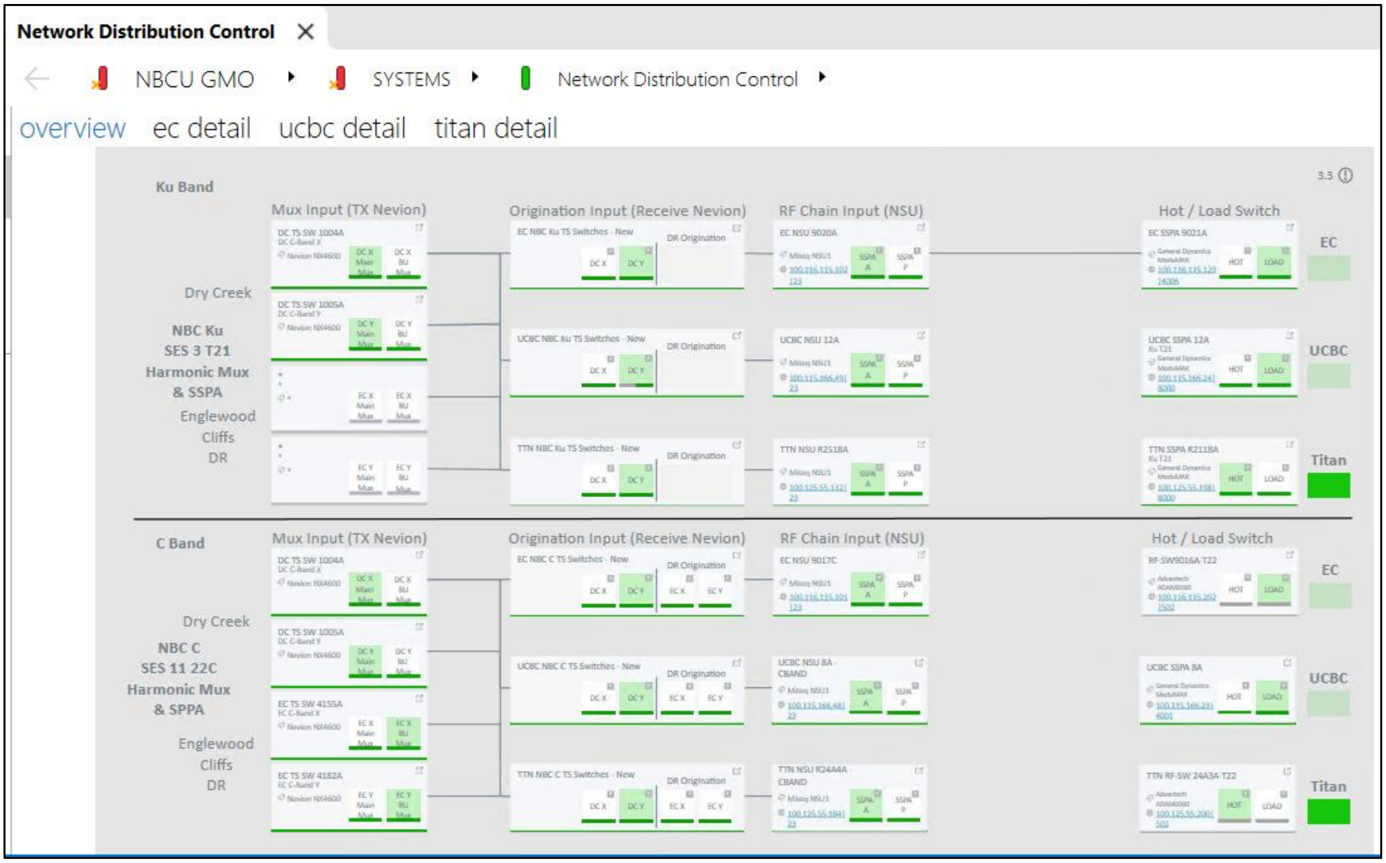


Figure 32 – DataMiner Network Distribution Control Overview Page

Satellite signals, like all RF signals, can be attenuated by severe weather or atmospheric conditions such as wildfire smoke. Having geo-diverse Earth Stations ensures that we have options. Ku-Band is more susceptible to attenuation than C-Band because it is a much higher frequency "Rain Fade" is a bigger concern, which is why we track weather forecasts and move Uplinks to avoid precipitation. Switches between sites are typically done during non-network time for Ku-Band and during local breaks on Telemundo for C-Band. This is done because there is a complete loss of signal at the receive sites while the active site is turned down to load and the new site is switched to the antenna.

Origination Sources

Each Earth Station has X/Y sources from Dry Creek Primary Origination, as well as X/Y EC DR origination for both C and Ku-Band Services. Both are monitored and managed using *DataMiner*.

TX Nevions: Both DC and EC have Main and Backup MUX for X and Y which are Live/Live meaning both active and streaming at the same time. Since we can only use one at a time, they are pulled into the *TX Nevions* which allows us to select whether main or backup is being sent to the WAN. This is the first selection or control for both Ku and C Band content presented in *DataMiner,* starting from left to right as *MUX Input* (see Figure 33, below). These will typically be on Main and will switch automatically to Backup (*BU*) if an error condition is detected. They can also be controlled directly by engineering when needed for maintenance and repairs. DC and EC each have X and Y TX Nevions and take the streams directly from the Harmonic Pro-Stream MUXs.

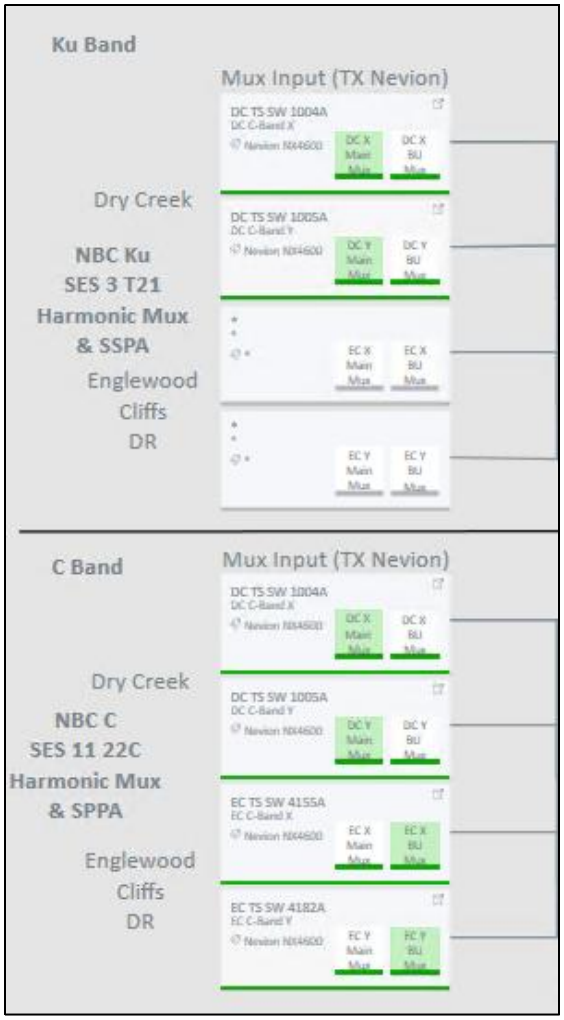


Figure 33 - DataMiner MUX Inputs (TX Nevions) for Ku and C band

RX Nevions: Each Uplink site has separate X and Y "RX Nevions" for C-Band and Ku-Band which connect to the WAN. Each one subscribes to all 4 Origination Sources for its chosen purpose, C-Band DC X/DC Y plus EC DR X/EC DR Y to all both X and Y C-Band units, and all 4 Ku-Band streams to Ku-Band units. This is what comprises the second column in DataMiner, "Origination Input" which is where you select which MUX Origination you want to transmit. When you select a source in this section of DataMiner it sets both A and Y units to output the same source as ASI to feed the next devices in the signal paths. Each site needs to be set individually (see , below). In other words, if you select DC-X for Ku T21 at Titan, only Titan will change, UCBC and EC will also need to be set. Making a change here will cause a glitch which is why origination switches are done in scheduled windows to avoid being seen by viewers however the glitch is minor compared to a site switch. In the event of a MUX failure this should be the first place to make a recovery switch.

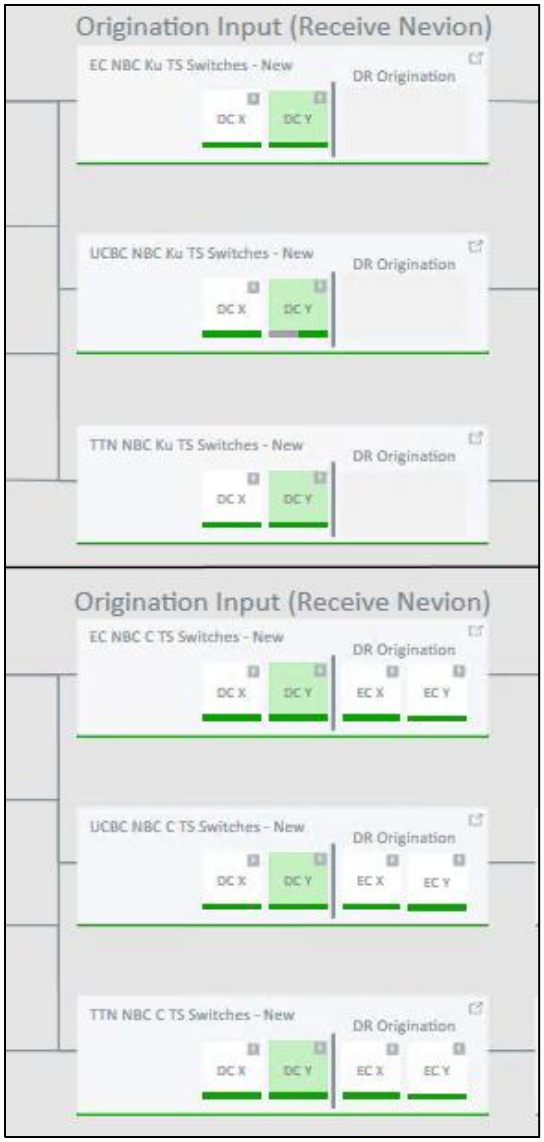


Figure 34 - Origination Input (Receive Nevion)

Uplink Signal Path

The following devices comprise the typical RF path and signal flow. Not all these devices are seen on the DataMiner Overview page but can be seen on the *Details* page for each site (Figure 35, below). The settings in these devices should only be accessed and manipulated by experienced engineering staff.

Figure 35 - DataMiner Titan Detail View

Modulators: Skypath utilizes X/Y redundant modulators that take ASI from the RX Nevions. The ASI is modulated into a Phase Shift Key signal and output as a 70MHz Modulated carrier wave. Our specification is DVB-S2 8PSK at 30MSym 5/6 FEC with .20 Rolloff and Pilots Active which occupies 36MHz of Bandwidth (see , below).

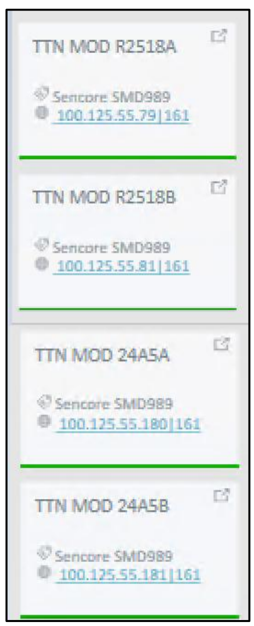


Figure 36 - DataMiner Titan Details tab. Column 2, Modulator info

Up-Converters: The X/Y 70MHz Signals are sent to Frequency Up-Converters (Figure 37, below) to bring them to “Native Uplink Frequency” which is 6365MHz for C-Band and 14420MHz for Ku-Band which are the center frequencies for the SkyPath Transponders.

Figure 37 - Up-Converter Info

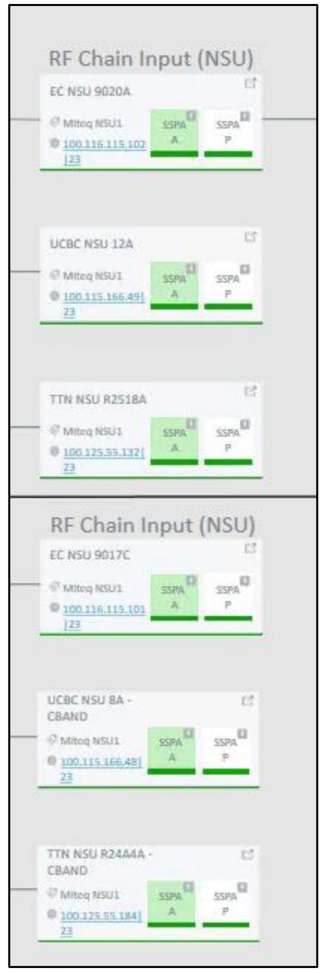


Figure 38 - RF Chain Input, NSU Info and control

Network Selector Unit Switch: The NSU Switch is a High Frequency RF Switch that selects between X/Y, or Air/Protect RF Signals (Figure 37, above). The NSU is also communicating with the Up-Converters to status and control them. This is where we converge to a single point at each site. Switching here is only done as an emergency recovery in the event of a Nevion, Modulator, or Up-Converter failure. For maintenance purposes the Y/Protect side might also be set by engineering.

Solid-State Power Amplifier (SSPA): The *SSPA* takes the output of the NSU, which is typically less than .1 SSPA watts and boosts it to 100 watts or more so that it can reach the satellite in orbit. The SSPA connects directly to the antenna feed and is the last device to touch the signals on their way to Affiliate receivers.

Hot/Load Switches: The Hot/Load Switch (Figure 39, following page) is the final control that allows the signal to the Antenna making it the active site. This allows the sites to be active and ready to speed recovery and keep RF levels stable. The equipment used for precision high frequency RF performs best when it is left running and “warmed up.” Some Hot/Load switches are waveguide switches that the SSPA controls, while others use a 50-ohm switch block upstream of the SSPA. Further details will be discussed later in this guide regarding the differences between sites.

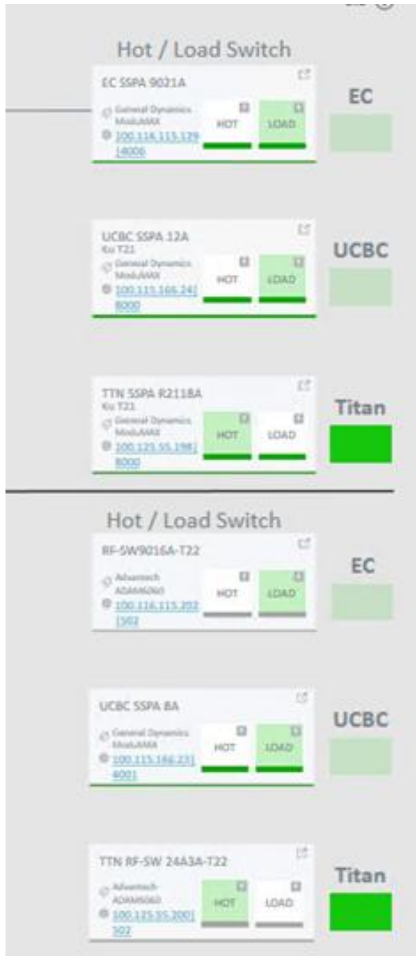


Figure 39 - Overview tab. Column 4 Hot/Load Switch Controls

TLT and Downlink Receivers: Test Loop Translators and Downlink signals are Harmonic PVR-7100 receivers used for monitoring the RF signals at various stages in the RF path. TLT's are autonomous devices that transpose frequencies from the Uplink of 14420MHz and 6365MHz to L-Band frequencies of 1370MHz and 1010MHz which are the same L-Band Frequencies of the Downlink return from the satellite that the receivers use. We also have receivers that take IP input from the TX Nevion outputs to monitor origination sources. Further details will be discussed in the monitoring section.

RF Redundancy and Failover

Each Earth Station uses X/Y redundant Nevion Virtuoso's for EACH service, C-Band and Ku-Band each have their own hardware as mentioned in the Sources section previously. This is also where the first recovery action should be taken if an Automatic Switch has not taken place. These are the bridges that bring the sources from the WAN /IP domain to Merge/Heal with SMPTE 2022-7, Select the source from all 4 options, and output ASI on a Coaxial Cable.

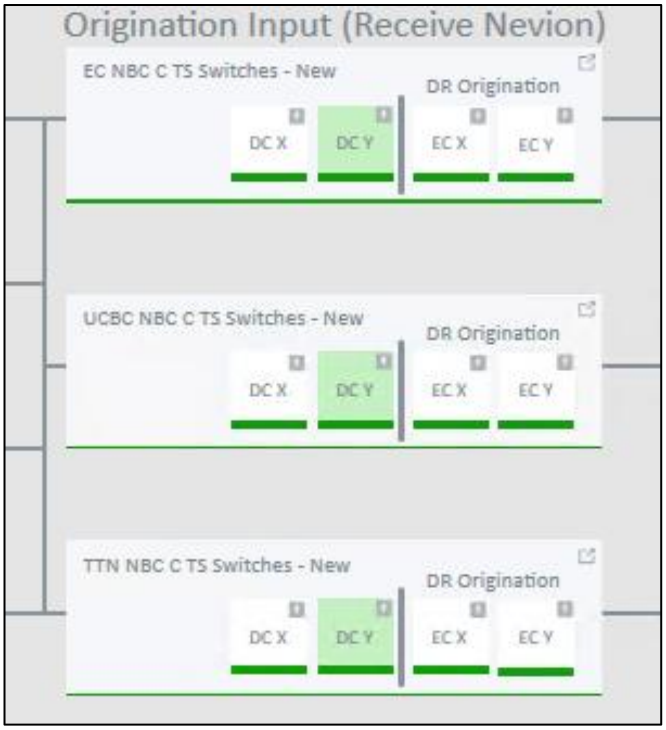


Figure 40 - Overview tab. Column 2, Origination Input

The green boxes in Figure 40, above, indicates the active source, typically DC X or DC Y. DC Y is Active in this example. The selection button controls both X/Air and Y/Protect Nevions to take the same source).

The next set of devices are the Modulators, and Up-Converters, discussed previously. They are also X/Y redundant but should also be regarded as Air/Protect (via the DataMiner Control Interface - Figure 41, on the following page).

The Network Selector Unit, or NSU is the switch point and control to select Air(X) or Protect(Y) RF Source to the SSPA. This is also the point at which redundancy wanes to singularity. There is only one NSU, and one SSPA per service, per location. They are connected directly to the antenna feed via Waveguide, a special copper tube to contain and convey the high-power RF.

All uplinks will typically be set to Air and the NSU left in Auto Fail-Over mode. There are, however, exceptions such as maintenance etc. A manual switch (**Error! Reference source not found.**, following p age) can be executed as well, in the event of a Nevion or Modulator failure that cannot be detected by the NSU.

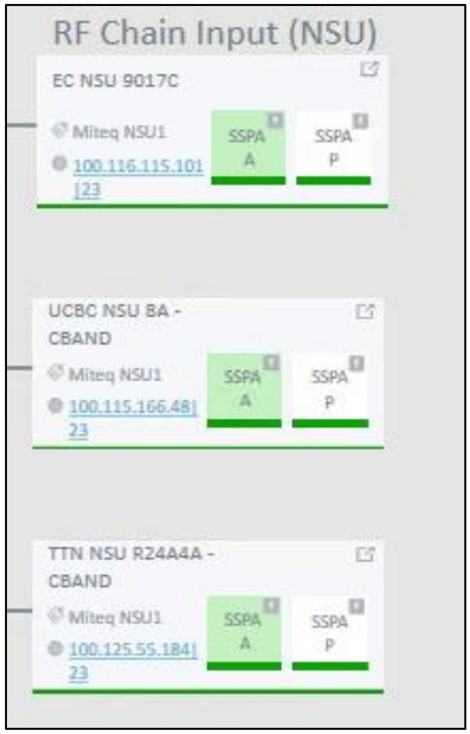


Figure 41 - C-Band NSU Air/Protect Control

Supplemental Info (Tools, Monitoring Points & Differences Between Sites)

This section briefly discusses a few tools as well as the various monitoring points of the Uplink paths. Monitoring points and differences between the sites are also be discussed.

Tools

Medius: Medius is an MPEG Transport Stream Analyzer and Video Decode quality logging tool. It has been used across the sites for many years and remains a useful tool.

Multiviewers: We use Multiviewers in the NOCs to monitor video, audio levels, and provide alarming based on quality of those signals. Layouts and UMD data help quickly identify problems to prompt quick recovery actions, or confirm an automatic action provided swift recovery.

Monitoring Points

MUX X/Y Origination: The first 2 columns of either carrier (C or Ku-Band) will be MUX X and MUX Y. These are taken the "TX Nevions" in Dry Creek and are the same that create the sources at each uplink. If the services in one of these columns freeze or drop the issue would likely be the origination MUX of the side that was impaired. If the active side failed that effects downlink an immediate X/Y Origination Source switch should be performed if an automatic switch did not engage.

RF Air/Protect: The RF Air and RF Protect signals originate from the Up-Converters at each uplink location, although it has not been fully implemented at time of this composition. The RF Test output is sent through a TLT and connected to an IRD. An issue seen here would indicate a problem with the side specific Nevion, Modulator, or Up-Converter at the given site. If either column went down on the active side, affecting Downlink, and an automatic switch did not occur, the NSU Switch should be manually engaged via Dataminer.

Downlinks: Downlinks are monitored at all 3 locations via IRDs connected to the Antenna's downlink feed. An interruption or disturbance at the active Uplink site will seen at all 3 sites, conversely a disturbance to the downlink feed at only one site is indicative of "Terrestrial Interference" or locally generated radio interference that can happen occasionally. Persistent interference should be reported to engineering for further investigation and troubleshooting.

Sat Returns: Sat Returns are IRDs that receive the satellite signal either through Downlink RF or via IP from another receiver connected to Downlink RF as described previously. These receivers output SDI video that can be routed to QC stations.

Differences Between Sites

Despite 'best efforts' it's not always possible to maintain uniformity across our sites. To this point the C-Band Uplinks in Englewood Cliffs NJ, and Titan operate slightly differently (e.g., SSPAs and Antennas transmit an additional C-Band Carrier on Transponder 16 for various NBCU Cable properties). To allow Network and Cable operations to remain independent of one another, switches and combiners were added upstream of the SSPA. Since Cable Uplinks do not use UCBC this feature was not needed and omitted. Also note that UCBC still has a News CUPS Uplink connected to the Ku Antennae. Also, neither EC nor Titan have CUPS capability.

Security – ZTA & IAM

As of this writing, NBCU employs Zero trust architecture (ZTA) as outlined in *NIST SP 800-207* in order to enforce least-privilege-per-request access in NBCUniversal and its Affiliate information systems and services. Zero trust architecture) utilizes zero trust concepts and encompasses component relationships, workflow planning, and access policies, and includes the network infrastructure (physical and virtual) and operational policies as part NBC Universal's ZTA plan.

Integral to NBCUniversal's implementation of ZTA is the integration of hub and spoke-level Identity and Access Management (IAM). NBCU's framework of business processes, policies, and technologies facilitates the management of electronic or digital identities, insuring that the right people and job roles in an organization can access the tools they need to do their jobs.

Detailed information on NBCU's implementation of ZTA and IAM is addressed in the document *NBCUniversal Cloud Architecture and Security*.

Revisions

Rev Level	Date	Changes
V0.1	11/25/22	First draft (partial).
V0.6	1/10/23	Current draft iteration